

CLAUDE CODE (GLM) SURVEY PIPELINE

THEORETICAL TOOLS IN DATABASE SYSTEMS RE- SEARCH

LECTURE NOTES DISTILLED FROM THE SIGMOD 2026 FULL-CORPUS SURVEY

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Introduction: What DB Researchers Prove, and How

What these notes are

These lecture notes are a *toolbox tour* of the theoretical machinery that database researchers actually use, extracted mechanically from the complete proceedings of **SIGMOD 2026**: 385 research papers (PACMMOD Vol. 4, rounds 3–4, plus Vol. 3 issues 4–6, rounds 1–2). Every paper was fetched (arXiv $\text{\LaTeX}$ sources when available, ACM camera-ready PDFs otherwise), machine-extracted into formal statements, proofs, and bound sentences (2,405 statements, 1,130 proofs), and 228 theory-relevant papers were tagged with a controlled vocabulary of properties, tools, domains, and roles. The aggregates behind every figure in this chapter come from that survey; every worked example in the later chapters is a real paper from the corpus, cited by its sigmod26-NNN key.

Two honesty markers used throughout:

- **PODS** marks exemplars from the **PODS'26 issue** (PACMMOD Vol. 4 No. 2, DOI band 10.1145/3801890–3801919): 29 of the 228 tagged papers (12.7%). They are part of the proceedings bundle and are kept because they are often the cleanest exhibits of a tool, but they carry the theory-first culture of a sibling conference.
- Formal statements are transcribed from machine extraction of the papers; where the extraction was noisy we paraphrase conservatively rather than decorate.

The survey questions

1. **Modeling** — how do papers formalize a systems problem (cost model, adversary, noise model, distance metric)?
2. **Properties** — what do they prove about the artifact: correctness, complexity bounds, approximation/competitive ratios, statistical guarantees?

3. **Tools** — which theoretical machinery delivers the proof?
4. **Role** — does theory motivate the work, drive the design, serve as the evaluation yardstick, or *is* it the contribution?

Properties: first correct, then fast

Figure 1 ranks what the 228 tagged papers prove. The headline surprise of the full-corpus count: **correctness is #1** (98 papers, 43%) — ahead of upper bounds (75) and probabilistic guarantees (42). The typical SIGMOD theory section is not a big theorem; it is a small invariant certifying that a component preserves a property (monotonicity of timestamps, order-independence of index construction, merge-safety of compressed blocks). Hardness (36) plays a different role: the standard *license for heuristics* in graph search, repair, and mining papers. The tail is where culture lives: **tightness** (11 — matching upper and lower bounds, a $5.5\times$ jump over the arXiv-only view of the corpus) and freshly activated slots such as competitive ratio and regret bounds (2 papers each, entering through systems doors: when to adapt, when to stop).

Tools: a trinity plus a riser

Figure 2 ranks the machinery. A *trinity* dominates: **reductions** (52 papers), **concentration inequalities** (39), and **induction** (36) — respectively the tools of the hardness culture, the statistical-certification culture, and the component-correctness culture. The big riser once the ACM-only mass arrived is **exchange-greedy** (4→17 papers): prove a selection problem is a submodular/metric instance, inherit the approximation factor. Chapter (the index table) maps each tool to its chapter.

Domains and dialects

Figure 3 shows where theory lives. Graph systems (56 papers) are now the dominant dialect — hardness → branch-and-bound/parameterized algorithms → index-correctness lemmas. Privacy (26) is a near-monoculture of composition accounting (69% of its papers), increasingly in Rényi/f-DP currency. Streaming/sketching splits into a pure-theory wing (matching lower bounds) and estimator systems; vector search (25) splits between CLT-style certification and structural correctness of graphs and routers.

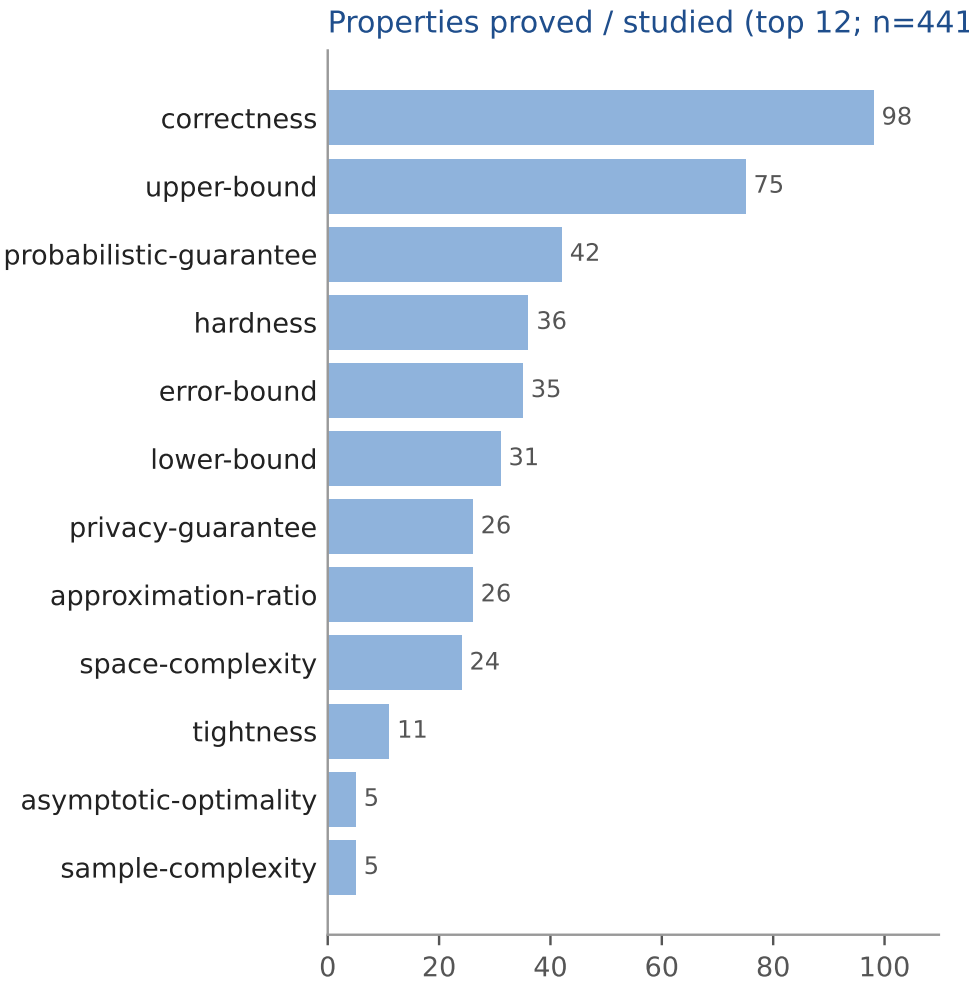


Figure 1: Most-studied properties across the 228 tagged papers.

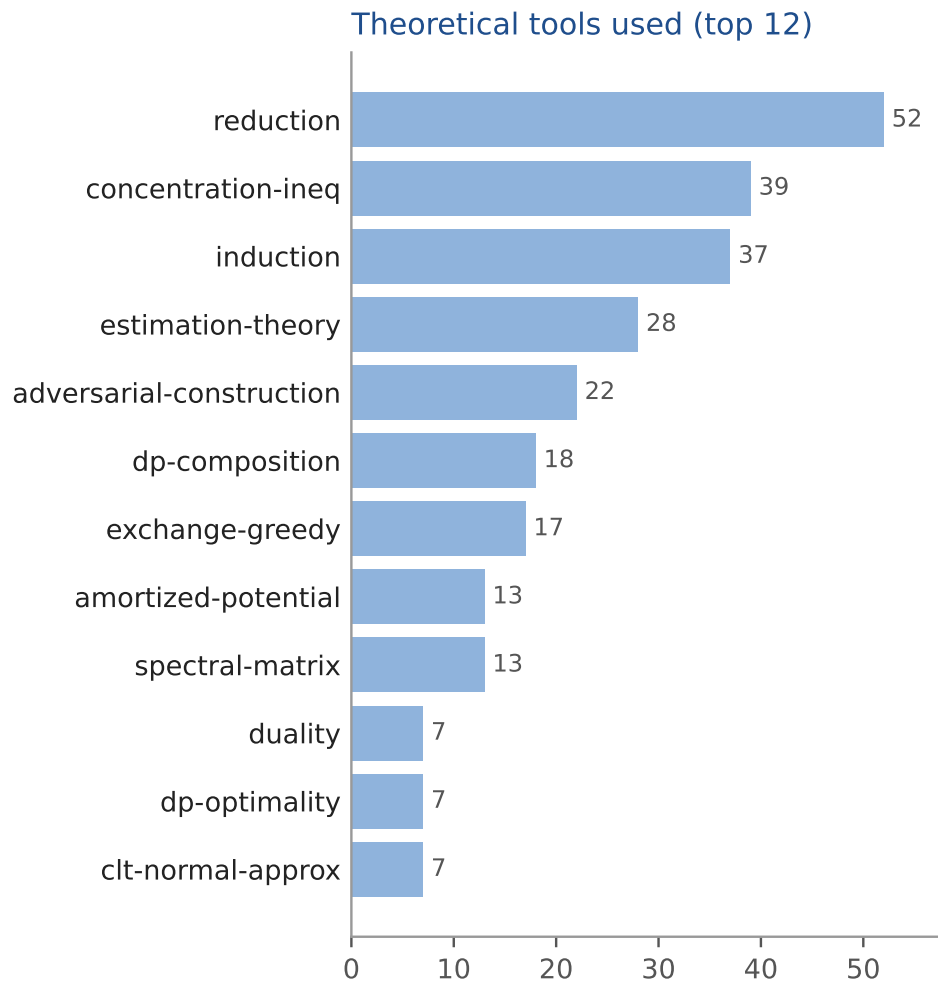
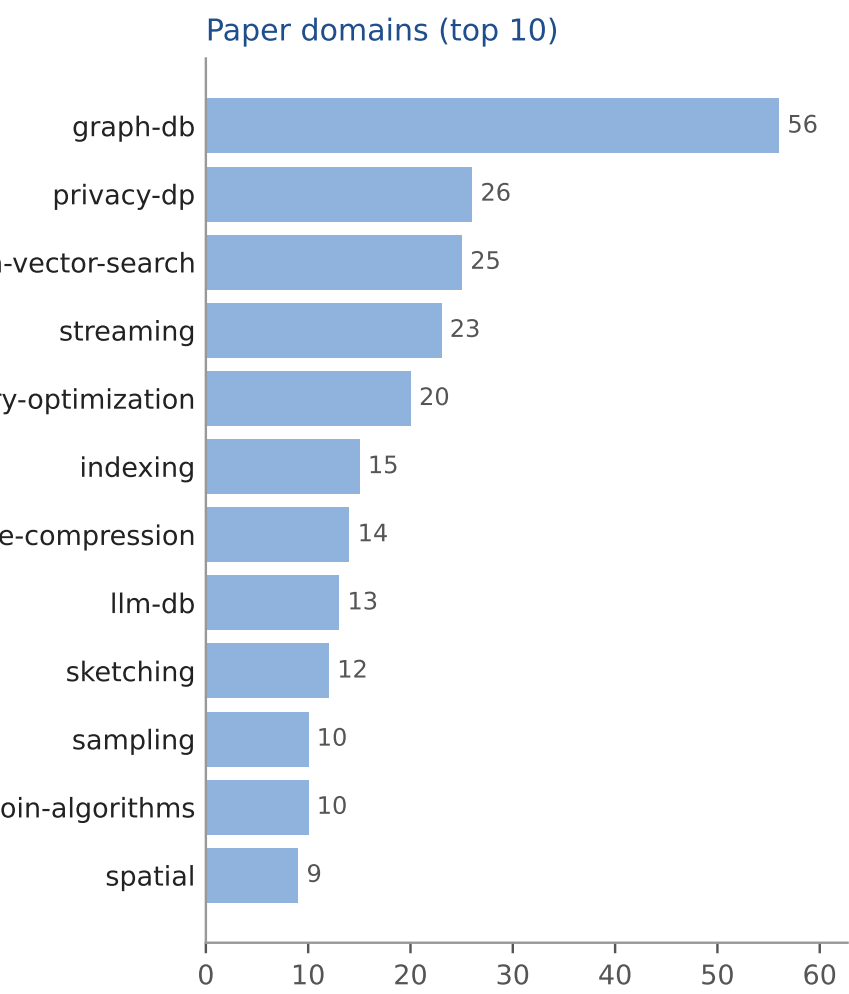


Figure 2: Most-used theoretical tools.
Each later chapter is one tool.



op 10).

The playbook: property $\times$ tool pairings

Figure 4 is the survey’s central cross-tab: which tool delivers which property. The two workhorses are **hardness $\times$ reduction** (32 papers) and **correctness $\times$ induction** (29). The pattern to internalize: DB theory compresses into about eight such pairings, and each chapter of these notes ends with the pairing(s) its tool owns.

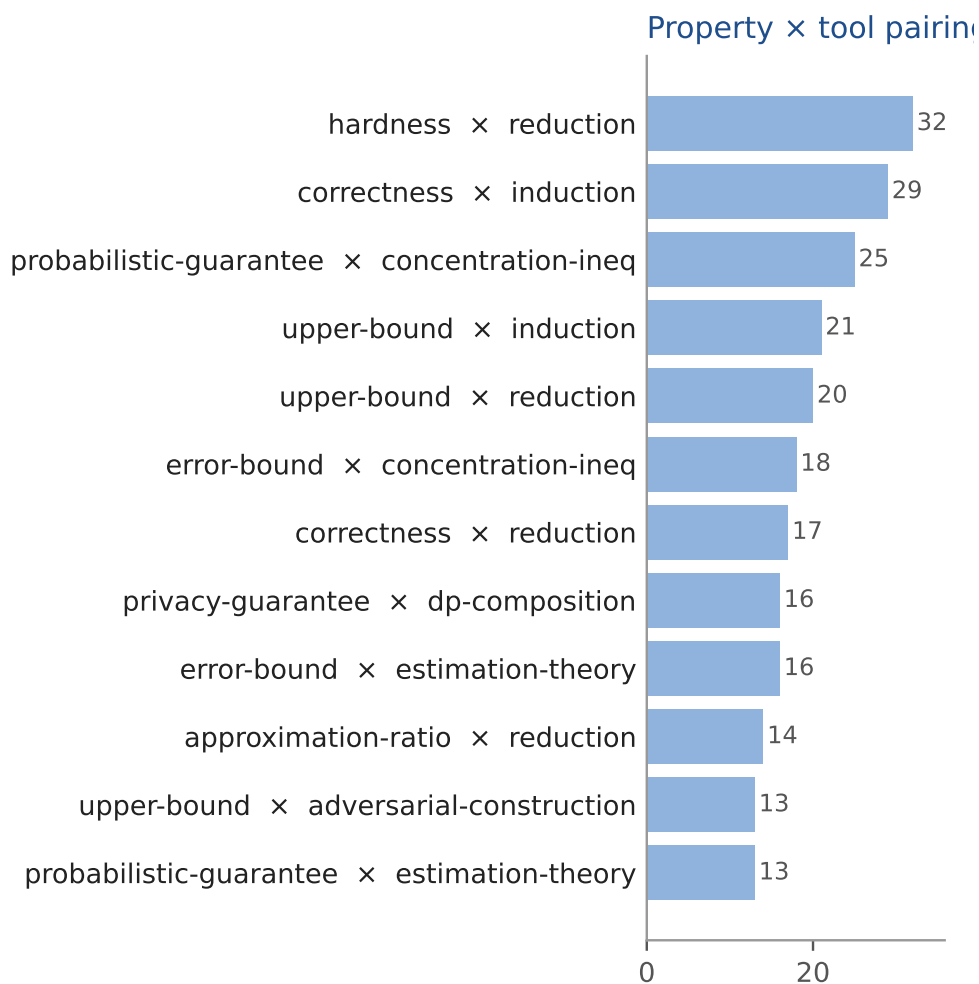


Figure 4: Top property $\times$ tool pairings — the “playbook”.

Where theory sits

66% of tagged papers use theory as a *design driver*; in 25% the formal result *is* the paper (concentrated in the PODS issue and the pure streaming enclave); motivation-only and evaluation-yardstick roles cover the rest. The role split matters when reading exemplars: a design-driver

paper proves what it builds; a theory-as-contribution paper builds nothing and proves everything.

Chapter map

Each of the following chapters is one tool: the formal core, then real SIGMOD 2026 exemplars with minimal context, then a “when to reach for it” box. Exemplars were selected by extraction score, deduplicated across chapters, and transcribed from the papers’ own statements; 44 of the 58 planned exemplars survived the authors’ diversity cuts. A ^{PODS} in the table marks chapters whose star exemplars include PODS-issue papers (cited with the same marker in the text).

	Papers	Ex.	Star exemplars
n inequalities	50	4	sigmod26-136 ^{PODS} , 289, 243 ^{PODS} , 109 ^{PODS}
	38	4	sigmod26-175 ^{PODS} , 104 ^{PODS} , 166 ^{PODS} , 196
	36	4	sigmod26-307, 371, 264, 011 ^{PODS}
theory	27	5	sigmod26-247 ^{PODS} , 012, 231, 373, 102
onstructions	21	5	sigmod26-004 ^{PODS} , 241 ^{PODS} , 244 ^{PODS} , 290, 306
on	18	5	sigmod26-298, 337, 263, 049 ^{PODS} , 073
edy arguments	16	5	sigmod26-350, 031, 053 ^{PODS} , 200, 009
otential analysis	11	5	sigmod26-363, 015, 274, 232 ^{PODS} , 141 ^{PODS}
atrix tools	13	3	sigmod26-303, 235, 253 ^{PODS}
ons (competitive & regret) ers” = tagged	4	4	sigmod26-242, 094, 221, 140

ol; “Ex.” = ex-
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Tools not large enough for their own chapter in this corpus but worth knowing by name: adversarial-adjacent duality/LP arguments (7 papers), dynamic programming optimality (7), CLT/normal approximations (7), dyadic peeling (6), sketch mergeability theory (6), information-theoretic lower bounds (5), balls-bins-hashing (5), coresets (5), communication complexity (4) — several appear as guest stars inside the ten chapters.

Caveats

The corpus mixes 137 arXiv-TeX extractions with 248 camera-ready PDF extractions (the latter noisier); the arXiv cohort is measurably theory-denser ($1.7\times$ extraction score), so depth-oriented shares carry that bias, and roughly a third of the apparent arXiv theory-friendliness was PODS contamination — hence the ^{PODS} markers. Tags come from a fixed vocabulary; long-tail techniques (branching recurrences, min-plus algebras, GPU work-span analysis) appear as free-text tags and are mentioned in chapters only when an exemplar uses them.

Concentration Inequalities

Learning Objectives

- Recognize when a system component needs a *with-high-probability* (whp) contract rather than an in-expectation one, and formalize the corresponding bad event.
- State — and be able to prove on a whiteboard — the ladder Markov $\rightarrow$ Chebyshev $\rightarrow$ Chernoff/Hoeffding $\rightarrow$ Azuma-Hoeffding $\rightarrow$ McDiarmid, and match each rung to the dependence structure of the problem at hand.
- Execute the standard recipe: tune each bad event's probability down to $\delta / (\text{number of events})$, union-bound over all copies of the component, and boost constant-probability guarantees by repetition with medians.
- Read a corpus paper's concentration argument critically: identify the random object, the independence/boundedness structure, the inequality used, and the δ budget.
- Understand why exponential tails (and not polynomial ones) are what make $\text{poly}(n)$ -way union bounds affordable.

Concentration inequalities bound the probability that a random quantity strays far from its expectation. For a systems paper they upgrade a randomized component that is correct *in expectation* — a sampler, sketch, random walk, or noise mechanism — into one correct *with high probability* (whp): a quantified contract such as “ $(1 \pm \epsilon)$ -accurate with probability $1 - 1/\text{poly}(n)$ ” or “running time $O(T)$ with probability $1 - 1/N$.” The recipe is constant: name the bad event, bound it by a tail inequality, union-bound over all copies of the component. In the SIGMOD’26 corpus this is among the most-used theoretical tools (38 tagged papers), directly supplying the probabilistic-guarantee property in 25 of them.

The tool: concentration inequalities

The problem class is uniform across applications: we can compute (or bound) the expectation of some random quantity X — the number of sampled points falling in a region, the error of a private estimate, the load of a hash bucket — but an algorithm must succeed on the *actual* draw, not the average. What concentration delivers is a bound of the form

$$\Pr[|X - \mathbb{E}X| \geq t] \leq \text{some decreasing function of } t,$$

plus the craft of making that function decrease fast enough. How fast is “fast enough” is the organizing question of this chapter: a bound of δ per bad event survives a union bound over $1/\delta$ events, so a component replicated $\text{poly}(n)$ times needs per-event failure $1/\text{poly}(n)$, which in turn needs tails that decay *exponentially* in the deviation.

The ladder

¹;;; and

Every bound in the family is Markov’s inequality in disguise ¹.

Theorem 0.1 — Markov. Let $X \geq 0$ be a random variable with $\mathbb{E}X < \infty$. Then for every $a > 0$,

$$\Pr[X \geq a] \leq \frac{\mathbb{E}X}{a}.$$

Proof. $\mathbb{E}X \geq \mathbb{E}[X \cdot \mathbf{1}\{X \geq a\}] \geq a \Pr[X \geq a]$; divide by a . $\square$

One moment, one side, polynomial tail. Applying it to $(X - \mu)^2$ instead of X gives the two-sided, second-moment rung:

Corollary 0.2 — Chebyshev. Let $\mu = \mathbb{E}X$ and $\text{Var}(X) < \infty$. Then for every $t > 0$,

$$\Pr[|X - \mu| \geq t] \leq \frac{\text{Var}(X)}{t^2}.$$

Proof idea. Markov applied to the nonnegative variable $(X - \mu)^2$ with $a = t^2$. $\square$

Polynomial tails suffice for constant success probability — boosted to high probability by repetition and medians — but they cannot survive a union bound over $\text{poly}(n)$ events. Exponential tails come from applying Markov to $e^{\lambda X}$:

Lemma 0.3 — Chernoff method. For any random variable X , any $a \in \mathbb{R}$ and any $\lambda > 0$,

$$\Pr[X \geq a] = \Pr[e^{\lambda X} \geq e^{\lambda a}] \leq e^{-\lambda a} \mathbb{E}[e^{\lambda X}],$$

and one optimizes over $\lambda > 0$ (the *moment generating function* controls the tail) ².

Evaluating the moment generating function of a sum of independent bounded variables yields the two workhorses of randomized systems. The multiplicative form is the one to memorize for sampling and sketching ³:

³;; and

Theorem 0.4 — Chernoff, multiplicative form. Let $X_1, \dots, X_m$ be independent random variables with $X_i \in [0, 1]$, $X = \sum_{i=1}^m X_i$, $\mu = \mathbb{E}X$. Then for every $\varepsilon \in (0, 1)$,

$$\Pr[X \geq (1 + \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/3}, \quad \Pr[X \leq (1 - \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/2}.$$

Proof idea. Chernoff method on X , then the elementary bound $(1 + \varepsilon) \ln(1 + \varepsilon) - \varepsilon \geq \varepsilon^2/3$ (and its lower-tail twin) on the optimized exponent; the full derivation is in Section . $\square$

The additive form is interchangeable whenever the summands are bounded but not naturally counts ⁴:

⁴;; and

Theorem 0.5 — Hoeffding, additive form. Let $X_1, \dots, X_m$ be independent with $X_i \in [a_i, b_i]$ a.s. and $X = \sum_i X_i$. Then for every $t > 0$,

$$\Pr[X - \mathbb{E}X \geq t] \leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right), \quad \Pr[X - \mathbb{E}X \leq -t] \leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right).$$

Proof idea. Chernoff method plus *Hoeffding's lemma*, $\mathbb{E}[e^{\lambda(X_i - \mathbb{E}X_i)}] \leq e^{\lambda^2(b_i - a_i)^2/8}$; optimize λ (proof in Section). $\square$

5

When independence fails but each random choice shifts the running statistic by a bounded amount, expose the randomness sequentially and book the shifts as a *martingale*⁵:

Theorem 0.6 — Azuma–Hoeffding. Let $(Y_i)_{i=1}^m$ be a martingale difference sequence with $|Y_i| \leq c_i$ a.s. Then for every $t > 0$,

$$\Pr\left[\sum_{i=1}^m Y_i \geq t\right] \leq \exp\left(-\frac{t^2}{2\sum_{i=1}^m c_i^2}\right).$$

Proof idea. Chernoff method on $\sum_i Y_i$, Hoeffding's lemma applied to each Y_i *conditionally on the past* (boundedness is all that is needed; independence is replaced by iterated conditioning), then optimize λ (Section). $\square$

The function-of-independent-variables form of Azuma–Hoeffding is the rung most used for adaptive algorithms and data structures, where the analyzed quantity is not a sum but a function whose per-variable influence is capped ⁷:

⁵ The entry fee for both forms is *independence plus bounded summands*. The tunable currency is the exponent $-\varepsilon^2\mu$ or $2t^2 / \sum_i (b_i - a_i)^2$ — which must be pushed past $\log(\#\text{events}/\delta)$ for the failure probability to amortize. When the summands are unbounded but light-tailed (Gaussian, Laplace, sub-exponential, sub-gamma), the same Chernoff-method machinery applies with a moment-generating function that exists near 0; corpus papers use sub-gamma tails for approximate counters (Morris-type) and Laplace tails for privacy noise, and we will see the latter in Section .

⁶ Recall: a martingale difference sequence (Y_i) with respect to a filtration $(\mathcal{F}_i)$ has $\mathbb{E}[Y_i | \mathcal{F}_{i-1}] = 0$; the partial sums $M_j = \sum_{i \leq j} Y_i$ then satisfy $\mathbb{E}[M_j | \mathcal{F}_{j-1}] = M_{j-1}$.

⁷;; and

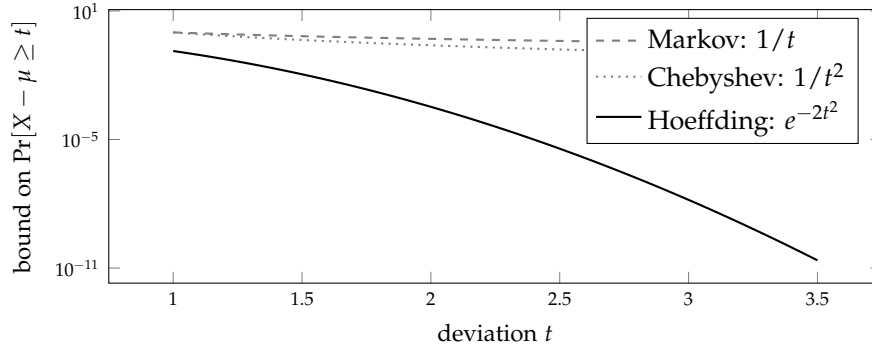


Figure 5: Tail decay for a unit-scale random quantity: moment bounds fall polynomially, independence buys the exponential curve — the gap that makes $\text{poly}(n)$ -way union bounds affordable.

Theorem 0.7 — McDiarmid, bounded differences. Let $X_1, \dots, X_m$ be independent random variables and $f: \prod_i \mathcal{X}_i \rightarrow \mathbb{R}$ a measurable function satisfying the bounded differences condition: for each i there is $c_i \geq 0$ with

$$|f(x_1, \dots, x_i, \dots, x_m) - f(x_1, \dots, x'_i, \dots, x_m)| \leq c_i$$

for all inputs and all x'_i . Then for every $t > 0$,

$$\Pr[f(X_1, \dots, X_m) - \mathbb{E}f \geq t] \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^m c_i^2}\right).$$

Proof idea. Apply Azuma–Hoeffding to the Doob martingale $Y_i = \mathbb{E}[f \mid X_1, \dots, X_i] - \mathbb{E}[f \mid X_1, \dots, X_{i-1}]$, whose increments are bounded by c_i exactly because changing X_i moves f by at most c_i (Section). $\square$

How to choose a rung

The selection procedure is mechanical once two questions are answered. *First, what depends on what?* If the random object is a sum of independent bounded indicators — samples retained, points in a region, hashed keys — use multiplicative Chernoff when the bound should be relative to the expectation μ (sampling rates, subsampling), and additive Hoeffding when it should be relative to the range parameters $b_i - a_i$ (weighted sums, real-valued measurements). If the algorithm is *adaptive* — later random choices conditioned on earlier estimates, sampling without replacement, stopping times — the summands are dependent but each step’s effect is bounded, and Azuma–Hoeffding applies. If the quantity is a *function* of many independent inputs (a data structure’s quality, a clustering cost, a query plan’s runtime) with per-input influence at most c_i , McDiarmid converts that directly. *Second, what moments are available?* If only the mean (or an exponent-optimized mean of a transform) is computable, Markov is the floor — and, applied to $e^{\lambda X}$, it is secretly the ceiling too. *Then tune:* set each bad

event's probability to $\delta/(\text{number of events})$, i.e. push the exponent past $\log(\text{number of events}/\delta)$, and union-bound.

When to Reach for Concentration

- Reach whenever a randomized component must be certified — sampling/sketching estimators, random walks, sampled data structures, hashing, privacy noise. The deliverable is a probability-qualified contract: accuracy $(1 \pm \epsilon)$ whp, or time/space bounds holding with probability $1 - 1/N$.
- Recipe: name the random object and the bad event; check dependence (Chernoff/Hoeffding for independent bounded summands, Azuma/bounded differences for adaptive exposure or capped per-variable influence); tune parameters so each bad event has probability at most $\delta/(\text{events})$; union-bound over all components; boost constant-probability guarantees by repetition with medians.
- Markov and Chebyshev suffice when only moments are available and constant success probability is enough — and Markov on $e^{\lambda X}$ is the engine of every exponential-tail argument anyway. Exponential tails are what make $\text{poly}(n)$ -way union bounds affordable; polynomial tails are not. Independence is the usual crux — hashing gives limited independence, adaptive algorithms create dependence — exactly where the martingale rung earns its keep.

Trimmed statistics in a turnstile stream

The paper cited in this section is a PODS'26 exemplar ^{8PODS} — kept because it is the cleanest concentration exhibit in the corpus: the entire correctness argument is one multiplicative Chernoff bound plus a union bound over logarithmically many levels.

8

Problem. Work in the *strict turnstile* streaming model: an n -dimensional frequency vector $\mathbf{x} \in \mathbb{Z}^n$ starts at zero and receives a long stream of updates of the form “add Δ to coordinate i ” (with Δ of either sign); the algorithm sees updates one at a time and must answer queries about the final $\mathbf{x}$ using space far less than n . Let $a_1, \dots, a_k$ be the k coordinates of $\mathbf{x}$ of largest absolute value. The *trimmed statistic* of order (p, k) is $\sum_{i=1}^k |a_i|^p$ for $0 \leq p \leq 2$: the F_p mass contributed by the top k coordinates — the part of the data that survives trimming away the tail. Such statistics are the streaming analogue of trimmed estimators in robust statistics.

Guarantee.

Theorem 0.8 — Top- k F_p estimation, Theorem 4 of the paper. Let $\mathbf{x} \in \mathbb{Z}^n$, $p \in [0, 2]$, $k \geq 0$, $\varepsilon \in (0, 1)$. Suppose $\|\mathbf{x}_{-k}\|_2^2 \geq (\varepsilon / \log n)^c \cdot k$ for a constant c , where $\mathbf{x}_{-k}$ excludes the top- k coordinates. Then there exists a *linear sketch* using $\text{poly}(\log n / \varepsilon)$ bits of space that estimates $\sum_{i=1}^k |a_i|^p$ to a $(1 \pm \varepsilon)$ factor with high constant probability.

9

⁹ A *linear sketch* stores $A\mathbf{x}$ for a random matrix A ; it is the standard route to mergeable, one-pass, deletion-tolerant stream summaries. The condition on $\|\mathbf{x}_{-k}\|_2^2$ holds automatically once $k \geq \text{polylog } n$, so the theorem covers the practically interesting regime.

¹⁰ CountSketch (Charikar–Chen–Farach-Colton) recovers, from hashed counters, all coordinates whose value exceeds a θ -fraction of the residual ℓ_2 norm (a *heavy hitter* structure); the paper needs only this black box.

How concentration is applied. The algorithm partitions coordinates by magnitude into *geometric level sets* whose boundaries are shifted by a random offset ζ uniform in $[1/2, 1]$, then *dyadically subsamples* each level: each coordinate of level S_j is kept independently with probability $r = \min\{1, C \log n / (\varepsilon^2 s_j)\}$ where $s_j = |S_j|$ and C is a constant. Survivors are found with a CountSketch¹⁰ fed with $\Theta(\log n)$ buckets per subsampling rate. The concentration step is the following lemma; its random objects are the independent retention indicators X_i of the coordinates of S_j , each Bernoulli(r), so the survivor count has mean $s_j r$.

Lemma 0.9 — Level-size estimation, Lemma 4.4 of the paper, notation cleaned. Let S_j be a level set with $s_j = |S_j|$, and subsample each coordinate independently with probability $r = \min\{1, C \log n / (\varepsilon^2 s_j)\}$. If z coordinates of S_j survive and $\tilde{s}_j = z/r$, then with probability $1 - 1/\text{poly}(n)$,

$$(1 - \varepsilon) s_j \leq \tilde{s}_j \leq (1 + \varepsilon) s_j.$$

Proof sketch. Write $u_1, \dots, u_{s_j}$ for the coordinates of S_j and $X_i = \mathbf{1}\{u_i \text{ sampled}\}$, so $\mathbb{E} \sum_i X_i = s_j r$. Multiplicative Chernoff gives $\Pr[|\sum_i X_i - s_j r| \geq \varepsilon s_j r] \leq 2 \exp(-\varepsilon^2 s_j r / 3) \leq 2 \exp(-C \log n / 3) \leq 1/\text{poly}(n)$; divide by r . (Full proof in Section.) $\square$

The δ tuning is explicit: r is chosen exactly so that the Chernoff exponent $\varepsilon^2 s_j r$ equals $C \log n$, i.e. per-level failure $1/\text{poly}(n)$; there are only $O(\log_{1+\varepsilon} m)$ levels (magnitude scales), so a union bound over levels preserves the polynomial failure probability. A companion lemma (Lemma 4.5 of the paper) uses the same Chernoff bound on the binomial survivor count to show that fewer than $O(\log n / \varepsilon^2)$ coordinates survive per level whp — which is what makes the heavy-hitter structure, and hence the space bound, $\text{poly}(\log n / \varepsilon)$.

What it buys. The system gets a one-pass, sublinear-space summary of trimmed statistics — robust aggregate mass that ignores the tail — with a certificate strong enough to union-bound into larger algorithms; the randomness of the offset ζ (rather than an adversarial grid) is what makes the level-set decomposition itself immune to worst-case input placement.

Maximum range sum by randomized grid shifting

Problem. In the *maximum range sum* (MaxRS) problem, the input is a set P of n points in $\mathbb{R}^d$, each carrying a weight, and a query shape Q — here a d -ball of given radius. The goal is to place Q (translate it anywhere) so as to maximize the total weight of the points of P covered. The paper cited here ¹¹ studies static, dynamic (point insertions/deletions), colored, and batched variants; spatial databases use MaxRS as the core subroutine behind “find the best location” queries.

¹¹

Exact MaxRS is expensive: for Q a 2-ball the colored version carries a conditional $\Omega(n^2)$ lower bound, and assuming that $(\min, +)$ -convolution on length- n sequences requires $\Omega(n^2)$ time, the batched variant on n points and m query radii requires $\Omega(mn)$ time — almost matching the trivial $O(mn \log n)$ upper bound.

¹²

Guarantee.

Theorem 0.10 — Static MaxRS, one representative case of the paper. For static MaxRS with Q a d -ball and any $0 < \varepsilon < 1/2$, there is a randomized $(\frac{1}{2} - \varepsilon)$ -approximation algorithm running in time $O(\varepsilon^{-2d-2} n \log n)$; the approximation factor holds with high probability, i.e. $1 - n^{-\Omega(1)}$. The dynamic version achieves the same factor with amortized update time $O(\varepsilon^{-2d-2} \log n)$.

¹² *Conditional lower bound:* a reduction from $(\min, +)$ -convolution — widely conjectured to need quadratic time — transfers that hardness to MaxRS. This is the paper’s evidence that *approximation* is necessary, and approximation is exactly where concentration enters.

How concentration is applied. The algorithm randomly shifts a grid over the point set and, per cell, takes a random sample of the cell’s points; the candidate balls it examines are drawn from a finite set of $k \leq n$ possibilities determined by the shift. Two facts combine. First, a *geometric covering lemma*: if C is a d -ball of radius ρ and B a unit d -ball whose center lies at distance ρ^2 from the center of C , then the $(d-1)$ -dimensional measure of the cap $B \cap \partial C$ is at least a $(\frac{1}{2} - \Theta(\varepsilon))$ fraction of the measure of ∂C — i.e., a random point of C ’s boundary lands in any fixed candidate ball with probability at least $\frac{1}{2} - \Theta(\varepsilon)$. Second, the concentration step:

Lemma 0.11 — Sampling concentration, Lemma of the paper, notation cleaned. Let $S = \{p_1, \dots, p_t\}$ be such that each point of S lies inside the candidate ball B independently with probability at least $\frac{1}{2} - \Theta(\varepsilon)$, and let $Y_B = \sum_{i=1}^t \mathbf{1}\{p_i \in B\}$. Then with high probability, simultaneously for all $k \leq n$ candidate balls,

$$Y_B \geq (1 - \varepsilon) \left(\frac{1}{2} - \Theta(\varepsilon) \right) t.$$

Proof sketch. The $X_i = \mathbf{1}\{p_i \in B\}$ are independent indicators with $\mathbb{E}[Y_B] \geq (\frac{1}{2} - \Theta(\varepsilon))t$; multiplicative Chernoff gives $\Pr[Y_B < (1 -$

$\varepsilon)\mathbb{E}Y_B] \leq e^{-\varepsilon^2 t/2}$, and a union bound over the $k \leq n$ balls keeps this whp provided $t = \Theta(\varepsilon^{-2} \log n)$. (Full proof in Section .) $\square$

The random objects are the per-point sampling/shift indicators; the structure is full independence across points (the random shift acts independently per cell); the rung is multiplicative Chernoff; and the δ tuning sets $t = \Theta(\varepsilon^{-2} \log n)$ so each of the n candidate balls fails with probability $n^{-\Omega(1)}$ — the same “push the exponent past $\log(\text{events})$ ” accounting as in Section , with the exponent coming from sample size rather than subsample rate.

What it buys. The spatial engine answers best-placement queries in near-linear time with a certified $(1/2 - \varepsilon)$ fraction of the optimum whp — and supports updates in polylog amortized time — where exact computation faces (conditional) quadratic barriers. The same Chernoff-plus-union-bound skeleton, re-instantiated with trapezoidal maps, gives the paper’s $(1 - \varepsilon)$ -approximation for the colored variant.

Private graph pattern counting under local differential privacy

Problem. Counting small patterns in a graph — how many length- k walks, or length- k simple paths, connect users — is a basic graph-analytics workload. This paper ¹³ must do it *under local differential privacy (LDP)*: there is no trusted curator, and each node of the graph randomizes its own messages before anything is aggregated. Two definitions make this precise. A randomized mechanism $M: \mathcal{G} \rightarrow \mathcal{K}$ satisfies ε -(edge-)differential privacy if for all edge-neighboring graphs G, G' (differing in one edge) and all measurable $S \subseteq \mathcal{K}$, $\Pr[M(G) \in S] \leq e^\varepsilon \Pr[M(G') \in S]$; the local variant additionally forces the randomization to sit at the nodes. A *walk* of length k may repeat vertices; a *path* may not — and W_k (respectively P_k) denotes the number of k -line walks (paths), with N the number of nodes and $d(G)$ the maximum degree.

Guarantee.

Theorem 0.12 — Accuracy of the k -round walk mechanism, Theorem 4.2 of the paper. For any graph G , walk length k , privacy budget ε , and failure parameter β , the mechanism $M^{k\text{-wk}}$ satisfies $\mathbb{E}[M^{k\text{-wk}}(G, \varepsilon)] = W_k$ and, with probability at least $1 - \beta$,

$$|M^{k\text{-wk}}(G, \varepsilon) - W_k| \leq k\gamma\sqrt{N} (d(G) + \gamma)^{k-1} = \tilde{O}\left(\sqrt{Nd(G)^{k-1}}\right), \quad \gamma = \frac{2k\sqrt{8\log(2/\beta)}}{\varepsilon}$$

the last equality for constant privacy budgets.

Since the number of k -walks is at most $Nd(G)^{k-1}$, the error scales as the square root of the maximal possible count — a meaningful

relative guarantee exactly on the sparse graphs that arise in practice. For simple paths, the paper reduces path counting to walk counting by *random marking* (each edge marked independently, only marked walks are counted); the resulting estimate is unbiased for P_k with error $O((k+1)\sqrt{N/\beta}(d(G)+k+1)^k)$, and this time the tail is only polynomial in $1/\beta$ (a variance/Markov argument), illustrating the price of dropping exponential tails.

How concentration is applied. The mechanism runs k rounds of message passing: node i 's round- ℓ value is the sum of its neighbors' round- $(\ell-1)$ values plus fresh Laplace noise of scale $2k\|X^{(\ell-1)}\|_\infty/\epsilon$, where $\|X^{(\ell-1)}\|_\infty$ is the largest absolute node value so far (privately calibrated *local sensitivity*). The random objects are the independent Laplace variables, one per node per round; their tails are *sub-exponential* ($\Pr[\text{Lap}(b) > ub] \leq \frac{1}{2}e^{-u}$), so the Chernoff method applies with the Laplace moment generating function. The δ -accounting is the chapter's recipe in its purest form: allocate failure budget β'/N per node per round, so that $2k\text{Lap}(\|X^{(\ell-1)}\|_\infty/\epsilon) \leq \gamma\|X^{(\ell-1)}\|_\infty$ holds for one node-round with probability $1 - \beta'/N$ (this is what fixes $\gamma = 2k\sqrt{8\log(2kN/\beta)}/\epsilon$); a union bound over the N nodes and k rounds makes the noise bound *simultaneous*; and then an *induction* on rounds — $\|X^{(\ell)}\|_\infty \leq (d(G) + \gamma)\|X^{(\ell-1)}\|_\infty$ — grows the noise scale geometrically but controllably, giving $(d(G) + \gamma)^{k-1}$. The final $\sqrt{N}$ is the standard $\sqrt{(\text{number of independent noise terms})}$ penalty when the N centered per-node Laplace contributions are summed. The full induction proof is transcribed in Section .

What it buys. The untrusted-aggregator setting gets pattern counts whose error depends on $\sqrt{N d(G)^{k-1}}$ rather than on the raw walk count's scale, with the privacy proof (Laplace sensitivity per round) and the utility proof (Laplace concentration per node per round) sharing the same random objects. The privacy–utility interplay is governed entirely by γ : more log factors in the failure budget translate linearly into noise scale.

The privacy of product noise as a concentration bound

Problem. Output perturbation is the simplest differential-privacy mechanism: release $f(x) + \text{noise}$. For $f: \mathcal{X} \rightarrow \mathbb{R}^M$ with ℓ_2 sensitivity $\Delta_2 f = \max\{\|f(x) - f(x')\|_2 : x, x' \text{ neighboring}\}$, the classical Gaussian mechanism adds $N(0, \sigma^2 I)$ with $\sigma = \Theta(\Delta_2 f \sqrt{\log(1/\delta)}/\epsilon)$. The paper¹⁴ instead *multiplies* the output by a product-structured random vector of scale σ_M and asks: when is the result (ϵ, δ) -differential private, i.e. $\Pr[M(x) \in S] \leq e^\epsilon \Pr[M(x') \in S] + \delta$ for neighboring x, x' ? The

analysis tool is the *privacy-loss random variable* (PLRV) $\mathcal{L}_M(s) = \ln \frac{dP}{dQ}(s)$, where P, Q are the output distributions on neighboring inputs: (ϵ, δ) -DP holds exactly when $\Pr[\mathcal{L}_M \geq \epsilon] \leq \delta$ — a tail bound in disguise.

Guarantee.

Theorem 0.13 — Product-noise DP, main theorem of the paper, compressed. Let $f: \mathcal{X} \rightarrow \mathbb{R}^M$ with $M > 2$ have ℓ_2 sensitivity $\Delta_2 f$, and let $k > 1$ be a constant. If

$$\sigma_M^2 \geq \frac{(\Delta_2 f)^2}{\epsilon^2} t^2, \quad t^2 = 2k^4 M \left(\frac{M}{4} + \frac{3}{2} \right)^{1+4/M} e^{1+2/M},$$

then the product noise of scale σ_M achieves (ϵ, δ) -DP on the perturbed $f(x)$, where δ is an explicit function of $\lambda = \Delta_2 f / \sigma_M$ involving confluent hypergeometric $({}_1F_1)$ terms. The expected magnitude of the product noise is smaller by a $\Theta(\log(1/\delta))$ factor than that of the classical Gaussian mechanism at the same privacy level.

How concentration is applied. The paper never writes the word “Chernoff,” but the argument is the Chernoff method verbatim, applied to a likelihood ratio rather than a sum of Bernoullis. First, the *product-measure lemma* (Radon–Nikodym derivatives multiply over products: $\frac{d(\nu_1 \times \nu_2)}{d(\mu_1 \times \mu_2)}(x, y) = \frac{d\nu_1}{d\mu_1}(x) \frac{d\nu_2}{d\mu_2}(y)$) factorizes the PLRV into independent per-coordinate terms — the independence structure. Each factor is a function of the angle between a random noise direction (uniform on the sphere S^{M-1}) and a fixed vector, whose density is a Beta-type law; its moments are computed in closed form. Second, the tail: the paper applies an *exponent-optimized Markov bound* — choose the power q , bound $\mathbb{E}[U^q]$ for the relevant transform U of the PLRV, and minimize the resulting $\delta = \delta_1 \delta_2$ over admissible q . That optimization over an exponent, applied to a nonnegative transform via Markov, is exactly Lemma 0.3 (the Chernoff method). Here the “ δ tuning” is literal: the DP parameter δ is the tail probability being bounded, and $\sigma_M \geq (\Delta_2 f / \epsilon) t$ is the noise-scale knob that drives it down.

What it buys. Because the noise’s radial and angular parts interact through the $\Theta(\log 1/\delta)$ -cheaper Gaussian-like tail, the mechanism adds less expected noise at equal privacy — and in the paper’s approximate-message-passing and convex-ERM experiments, accuracy at fixed privacy consistently improves, occasionally exceeding non-private baselines on specific datasets.

User-level private query processing with sampled tuples

Problem. Consider a select–project–join–aggregate workload over a relational instance whose rows belong to *users*, each user contributing many tuples. *User-level DP* demands the definition of Section with “neighboring” meaning *all* tuples of one user added or removed — much stricter than tuple-level DP. Unbounded per-user contributions then blow up sensitivity, so the system¹⁵ truncates each user’s contribution to a threshold τ chosen privately by a search (sparse vector technique), and accelerates the search by Poisson-sampling tuples at rate q before privatizing.

¹⁶

Guarantee.

Theorem 0.14 — Error of the sampled truncation, Algorithm 2 of the paper, compressed. The mechanism satisfies $(\epsilon, 0)$ -DP. Let n be the number of records, m the number of users, q the sample rate, $\tau^*(T) = \max_u \sum_{t \in T} \mathbf{1}[u \prec t]$ the true maximum per-user contribution, and $\epsilon_i^* \geq \epsilon/L$ the noise-calibration budget over the L search iterations. With probability $1 - \beta$, the released answer $\hat{f}(T)$ obeys

$$|\hat{f}(T) - f(T)| = \tilde{O}\left(\sqrt{\frac{n}{q}} + \frac{1}{q} + \frac{1}{\epsilon_i^*} \left(\tau^*(T) + \sqrt{\frac{\tau^*(T)}{q}}\right)\right),$$

where $\tilde{O}(\cdot)$ hides constants and polylogarithmic factors in m and $1/\beta$.

How concentration is applied. The random objects are the independent sampling indicators $\mathbf{1}\{t \text{ sampled}\} \sim \text{Bernoulli}(q)$ over the n tuples. The sampled statistic $f(S) = \sum_{t \in T} w_t \mathbf{1}[t \text{ sampled}]$ satisfies $\mathbb{E}[f(S)] = q f(T)$ and $\text{Var}[f(S)] = q(1-q) \sum_t w_t^2 \leq qn$: a sum of independent bounded variables with an explicit variance proxy — the territory of the Bernstein/Chernoff rung, whose bound gives $|f(S) - qf(T)| \lesssim \sqrt{qn \log(3L/\beta)}$ with probability $1 - \beta/(3L)$; dividing by q yields the $\sqrt{n/q}$ term. The δ -budgeting is again structural: the truncation-threshold search runs L iterations, each is allocated failure probability $\beta/(3L)$, and a union bound over iterations upgrades the per-iteration bound to the global $1 - \beta$ contract — with the same per-event log factor $\log(3L/\beta)$ appearing inside the noise scale. The $\tau^*(T)/\epsilon_i^*$ terms come from the Laplace-style calibration of the final release against the (privately learned) per-user cap.

What it buys. The query engine can offer user-level — not merely tuple-level — privacy at a modest accuracy price: the sampling error $\sqrt{n/q}$ decouples from the privacy noise, so tuning q trades compute for

¹⁵

¹⁶ Poisson sampling also *amplifies* privacy: a mechanism that is ϵ -DP on the sample is (ϵ') -DP on the full data with $\epsilon' = \ln(1 + q(e^\epsilon - 1)) < \epsilon$; the paper’s Sample-and-Explore comparison baselines use this. The proof is a binomial tail computation — again concentration.

accuracy without touching the privacy guarantee, and the explicit $\tau^*(T)$ dependence quantifies exactly how much heavy user-contributors cost the answer.

Deferred proofs

Chernoff, multiplicative form (Theorem 0.4). By the Chernoff method, for $\lambda > 0$, $\Pr[X \geq (1 + \varepsilon)\mu] \leq e^{-\lambda(1+\varepsilon)\mu} \prod_i \mathbb{E}[e^{\lambda X_i}]$. For $X_i \in [0, 1]$ independence gives $\mathbb{E}[e^{\lambda X_i}] = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1))$, so the product is at most $e^{\mu(e^\lambda - 1)}$ and, choosing $\lambda = \ln(1 + \varepsilon)$,

$$\Pr[X \geq (1 + \varepsilon)\mu] \leq \left(\frac{e^\varepsilon}{(1 + \varepsilon)^{1+\varepsilon}} \right)^\mu \leq e^{-\varepsilon^2 \mu / 3},$$

using $(1 + \varepsilon) \ln(1 + \varepsilon) - \varepsilon \geq \varepsilon^2/3$ for $\varepsilon \in (0, 1)$. The lower tail is symmetric, with exponent $\varepsilon^2/2$.

Hoeffding, additive form (Theorem 0.5). *Hoeffding's lemma:* if $Y \in [a, b]$ a.s. and $\mathbb{E}Y = 0$, then $\mathbb{E}[e^{\lambda Y}] \leq e^{\lambda^2(b-a)^2/8}$. (Proof: $e^{\lambda y}$ is convex, so $e^{\lambda y} \leq \frac{b-y}{b-a}e^{\lambda a} + \frac{y-a}{b-a}e^{\lambda b}$ pointwise; take expectations and minimize over a shifted λ .) Applying the lemma to each centered summand and multiplying by independence, $\mathbb{E}[e^{\lambda(X - \mathbb{E}X)}] \leq e^{\lambda^2 \sum_i (b_i - a_i)^2/8}$; the Chernoff method then gives $\Pr[X - \mathbb{E}X \geq t] \leq e^{-\lambda t + \lambda^2 \sum_i (b_i - a_i)^2/8}$, and optimizing at $\lambda = 4t / \sum_i (b_i - a_i)^2$ yields the stated $e^{-2t^2 / \sum_i (b_i - a_i)^2}$.

Azuma-Hoeffding and McDiarmid. Let $M_j = \sum_{i \leq j} Y_i$ with $\mathbb{E}[Y_i | \mathcal{F}_{i-1}] = 0$ and $|Y_i| \leq c_i$. Iterated conditioning gives $\mathbb{E}[e^{\lambda M_m}] = \mathbb{E}[e^{\lambda M_{m-1}} \mathbb{E}[e^{\lambda Y_m} | \mathcal{F}_{m-1}]]$, and conditionally on $\mathcal{F}_{m-1}$ the variable Y_m is centered and bounded in an interval of length $2c_m$, so Hoeffding's lemma applies with $(b - a) = 2c_m$. Unrolling, $\mathbb{E}[e^{\lambda M_m}] \leq e^{\lambda^2 \sum_i c_i^2/2}$, and optimizing λ gives the stated tail. For McDiarmid, set $Y_i = \mathbb{E}[f | X_1, \dots, X_i] - \mathbb{E}[f | X_1, \dots, X_{i-1}]$: this is a martingale difference sequence with $|Y_i| \leq c_i$ precisely because resampling only X_i moves f by at most c_i (take conditional expectations over the two values of X_i), and $\sum_i Y_i = f - \mathbb{E}f$. Azuma-Hoeffding applied to (Y_i) is McDiarmid's inequality.

Level-size estimation in the turnstile stream (Lemma of Section). Denote the coordinates of S_j by u_i , $i \in [s_j]$, and let X_i be the indicator that u_i is sampled into substream P , so the X_i are independent with $\mathbb{E}[X_i] = r$ and $\mathbb{E} \sum_i X_i = s_j r$. Multiplicative Chernoff (Theorem 0.4) gives

$$\Pr\left[\left|\sum_i X_i - s_j r\right| \geq \varepsilon s_j r\right] \leq 2 \exp(-\varepsilon^2 s_j r / 3) \leq 2 \exp(-C \log n / 3) \leq 1 / \text{poly}(n),$$

where the second inequality uses $s_j r = C \log n / \varepsilon^2$ (when the minimum in the definition of r is attained at this branch; otherwise $r = 1$ and

there is nothing to estimate). Dividing by r converts the event into $(1 - \varepsilon)s_j \leq \tilde{s}_j \leq (1 + \varepsilon)s_j$.

Sampling concentration for MaxRS (Lemma of Section). Fix a candidate ball B and let $X_i = \mathbf{1}\{p_i \in B\}$; the X_i are independent, and by the covering property $\mathbb{E}[Y_B] = \sum_i \mathbb{E}X_i \geq (\frac{1}{2} - \Theta(\varepsilon))t$. Multiplicative Chernoff gives

$$\Pr[Y_B < (1 - \varepsilon)\mathbb{E}Y_B] \leq e^{-\varepsilon^2 \mathbb{E}Y_B/2} \leq e^{-\Theta(\varepsilon^2 t)}.$$

With $t = \Theta(\varepsilon^{-2} \log n)$ this is $n^{-\Omega(1)}$ per ball, and a union bound over the $k \leq n$ candidate balls gives the simultaneous statement $Y_B \geq (1 - \varepsilon)(\frac{1}{2} - \Theta(\varepsilon))t$ for all of them, with high probability.

Round-by-round noise control in the LDP walk mechanism (Theorem of Section). Write $X_i^{(\ell)}$ for node i 's round- ℓ value. The mechanism gives, for each node i and round $\ell \in \{1, \dots, k\}$,

$$X_i^{(\ell)} = \sum_{j \in N(i)} X_j^{(\ell-1)} + \text{Lap}(2k \|X^{(\ell-1)}\|_\infty / \varepsilon),$$

so $\sum_{j \in N(i)} X_j^{(\ell-1)} \leq d(G) \|X^{(\ell-1)}\|_\infty$ always. Allocate failure β'/N per node per round: the Laplace tail $\Pr[\text{Lap}(b) > u] \leq \frac{1}{2}e^{-u/b}$ gives, with probability $1 - \beta'/N$ per node-round, $2k \text{Lap}(\|X^{(\ell-1)}\|_\infty / \varepsilon) \leq \gamma \|X^{(\ell-1)}\|_\infty$ for the stated γ (with $\beta' \approx \beta/k$). Union-bounding over the $\leq kN$ node-rounds, all these events hold simultaneously with probability $\geq 1 - (k-1)\beta'$, and on that event an induction gives

$$\|X^{(\ell)}\|_\infty \leq (d(G) + \gamma) \|X^{(\ell-1)}\|_\infty \implies \|X^{(\ell)}\|_\infty \leq (d(G) + \gamma)^\ell$$

for all $\ell \leq k-1$. The released count aggregates the N per-node values, whose centered Laplace contributions are independent sub-exponential variables each of scale at most $O(\gamma(d(G) + \gamma)^{k-1})$; their sum concentrates around zero at scale $\sqrt{N} \cdot \tilde{O}(\gamma(d(G) + \gamma)^{k-1})$ by the same Chernoff-method argument applied to light-tailed variables, which yields the stated $k\gamma\sqrt{N}(d(G) + \gamma)^{k-1}$ error radius with probability $1 - \beta$.

Chapter Recap

- **What the tool is.** Concentration inequalities convert expectations into with-high-probability contracts: $\Pr[|X - \mathbb{E}X| \geq t]$ bounded by a tail function, tuned so that per-event failure $\delta/(\text{events})$ survives a union bound over all copies of the randomized component.
- **Which rung when.** Independent bounded indicators/sums: multiplicative Chernoff (relative error) or additive Hoeffding

(range-based). Adaptive or dependent exposure with bounded per-step effect: Azuma–Hoeffding. Function of many independent inputs with capped per-input influence: McDiarmid. Only moments available: Markov and Chebyshev — constant-probability results, boostable by medians. Light-tailed unbounded noise (Laplace, Gaussian, sub-gamma): the Chernoff method still applies via the moment generating function.

- **Why exponential tails.** Only $e^{-\Omega(\text{exponent})}$ tails make $\text{poly}(n)$ -way union bounds affordable; polynomial tails ($1/t$, $1/t^2$) cannot amortize — visible in the corpus as the gap between $\tilde{O}(\cdot)$ guarantees (Laplace/Chernoff paths) and $1/\beta$ -type errors (variance-only paths).
- **What the corpus showed.** Trimmed-statistic sketching ^a]: Chernoff on subsampling survivors, union over $O(\log_{1+\varepsilon} m)$ levels. Spatial MaxRS ^b]: Chernoff on sampled point-in-ball indicators, union over $\leq n$ candidate balls. LDP graph counting ^c]: per-node-per-round β'/N budgets, union over kN node-rounds, then induction on rounds. Product-noise privacy ^d]: Markov at an optimized exponent on the privacy-loss random variable — the Chernoff method on a likelihood ratio. User-level private queries ^e]: Bernstein/Chernoff on Bernoulli-sampled sums with $\beta/(3L)$ budgets over search iterations.
- **The recipe, once more.** Name the random object; check independence or bounded differences; pick the rung; push the exponent past $\log(\text{events}/\delta)$; union-bound; boost by repetition if only constant probability was achievable.

^a [

^b [

^c [

^d [

^e [

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Reductions: Importing Hardness into Database Problems

Most “impossible” statements in database research are not proved from scratch: they are *imported*. A reduction exhibits a computable translation from a problem already believed hard — CLIQUE, Set Cover, inner-product estimation, hyperclique detection — into the problem at hand, so that any algorithm, approximation, private mechanism, or sampler for the target would also solve the source. Reductions are the most used theoretical tool in our SIGMOD’26 corpus: 50 of 228 tagged papers rely on them, and 32 of those pair a reduction with an explicit hardness claim. They matter in both directions: they delimit what any system can do (lower bounds, dichotomies, inapproximability) and, run backwards, they turn new problems into old ones with known algorithms.

The tool: mappings that preserve what you care about

Definition 0.15 (Polynomial-time many-one reduction). Let L_1, L_2 be decision problems. We write $L_1 \leq_m^p L_2$ if there is a polynomial-time computable function f with $x \in L_1 \iff f(x) \in L_2$ for all x . If some NP-hard L_1 reduces to L_2 , then L_2 is NP-hard: a polynomial-time algorithm for L_2 , composed with f , would decide L_1 .

This is the classical core ¹⁷, and it already reaches into databases.

¹⁷;; and

Example 0.16 (Conjunctive-query evaluation is NP-complete). Given (G, k) , build in polynomial time the database D_G holding the edge relation of G , and let Q_k be the conjunctive query

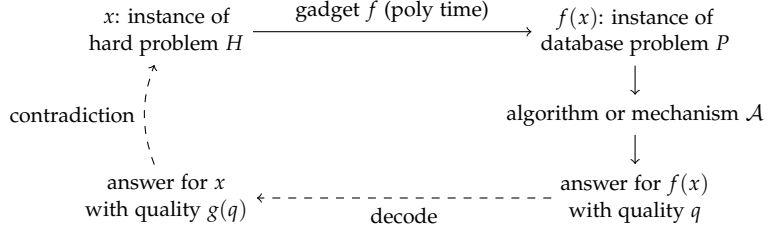
$$Q_k = \exists v_1 \dots \exists v_k. \bigwedge_{1 \leq i < j \leq k} E(v_i, v_j).$$

Then G has a k -clique iff $Q_k(D_G) \neq \emptyset$. Since CLIQUE is NP-complete, CQ evaluation is NP-hard (and it lies in NP) — the starting point of query-evaluation complexity ¹⁸.

¹⁸

Decision-hardness is often not enough: DB problems carry objectives

Figure 6: Every reduction has two halves: instances travel forward through the gadget; solutions — or accuracy, or privacy — travel backward, so quality achieved for P becomes quality for H .



(cost, error, latency), so the reduction must preserve *quality*, not just yes/no membership.

Definition 0.17 (*L-reduction*, i.e., approximation-preserving reduction). Optimization problem A *L-reduces* to B if there are polynomial-time maps f (instances) and g (solutions) and constants $\alpha, \beta > 0$ such that, for every instance x of A , writing c_A, c_B for the objective values: (i) $\text{OPT}_B(f(x)) \leq \alpha \cdot \text{OPT}_A(x)$; and (ii) every solution y of $f(x)$ maps to a solution $g(y)$ of x with $|c_A(g(y)) - \text{OPT}_A(x)| \leq \beta \cdot |c_B(y) - \text{OPT}_B(f(x))|$.

Proposition 0.18 If A *L-reduces* to B with constants α, β and B admits a polynomial-time r -approximation, then A admits an $\alpha\beta r$ -approximation. Hence hardness of *approximating* A — e.g., Set Cover’s $\log n$ threshold under the Unique Games Conjecture — transfers to B ¹⁹.

Remark 0.19 (Fine-grained and conditional reductions). When polynomial versus exponential time is too coarse, the source problem H carries a quantitative hypothesis (SETH, combinatorial Boolean matrix multiplication, hyperclique detection, listing lower bounds) ²⁰. The reduction must then track parameters: an algorithm for P with cost c_P on instances produced by f yields an algorithm for H of cost (cost of f) + (number of calls) $\cdot c_P$, and the hypothesis transfers whenever this beats the assumed lower bound in every parameter regime.

The same skeleton covers quantitative guarantees: replace “algorithm” by “mechanism or estimator with error α ,” and the reduction lifts accuracy instead of time. Figure 6 schematizes the two halves.

Reductions at work in the SIGMOD’26 corpus

Gadget reductions: privacy under continual release

A stream of edge updates arrives over T timesteps, and an (ϵ, δ) -differentially private mechanism must continually report, on n -node graphs, the size of the maximum matching, the degree histogram, or the core number of a vertex.

Theorem 0.20 — Continual-release lower bound. Let $\epsilon \in [0, 1]$ and $\delta \in [0, 1/3]$.

¹⁹; and

²⁰

Any (ϵ, δ) -DP incremental mechanism in the exact setting for maintaining the size of the maximum matching, the degree histogram, or the core number of a vertex must have additive error $\alpha = \Omega(\min\{T^a, n^b\})$, for problem-specific constants $a, b \in [1/8, 1/2]$.

The reduction embeds an INNERPRODUCT instance of dimension d into the update stream via *extendable bitwise-AND gadgets* with $v(d)$ vertices and $t(d)$ edges (for matching, $v(d) = t(d) = O(d^2)$): a mechanism with additive error α would answer the inner product within 2α , forcing $\alpha = \Omega(\min\{t^-(T), v^-(n)\})$ for generalized inverses, with d chosen exactly as that minimum. Binary-tree mechanisms with DP composition give near-matching upper bounds, so the reduction is essentially tight. ^{21PODS}

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Gap reductions: clustering with set outliers

Points are clustered while whole *sets* may be discarded as outliers: given points P , a family $\mathcal{H}$ of candidate outlier sets (hyper-rectangles, or sets induced by a relational join), and budgets k, z , choose at most k centers and at most z outlier sets so as to minimize the cost of the uncovered points.

Theorem 0.21 — Inapproximability under the Unique Games Conjecture. If the Unique Games Conjecture is true, there exists no $(1, f - \epsilon, \gamma)$ -approximation algorithm that runs in polynomial time for the problem, whenever $f = o(\log n)$, where the triple bounds the clustering-cost factor and the allowed violation of the center and outlier budgets.

The reduction is a line-point encoding of Set Cover: element x_i becomes the point p_i at coordinate i , each set y becomes the outlier set $\{p_i : x_i \in y\}$, and singleton sets pad the family, so covering the element-points with at most z outlier sets is exactly a set cover — and Set Cover’s UGC-conditional $\log n$ threshold crosses over. The same paper runs the machinery constructively: LP feasibility with greedy rounding, weight-charging, and coresets yield $(2 + \epsilon, 2, O(1))$ -approximations, including over acyclic joins. ²²

22

Coupon-collector reductions: sampling join-project queries

For a join-project query Q (a join followed by projection, so duplicates disappear), with input size N and output size OUT , the paper asks what uniform sampling from, and approximate counting of, $Q(I)$ must cost after indexing.

Theorem 0.22 — Conditional bound for star queries. For the star query Q_k , assuming the combinatorial hardness of listing star-join results (any

combinatorial listing algorithm requires $\Omega(N \cdot \text{OUT}^{1-1/k})$ time, under the property-testing model), no combinatorial algorithm can preprocess an instance of input size N and output size OUT in $O(N \cdot \text{OUT}^{1/k-\gamma})$ time and then return a uniform sample from $Q(I)$ in $O(\text{OUT}^{1/k+\gamma})$ expected time, for any constant $\gamma > 0$.

The reduction runs the coupon collector backwards: a sampler this fast would list all OUT results after $\text{OUT} \log \text{OUT}$ draws and thereby beat the assumed listing lower bound. For the matrix query the hypothesis is instantiated by the combinatorial hardness of Boolean matrix multiplication (the case $k = 2$), and any randomized algorithm that outputs an $O(1)$ -approximation of OUT with high probability requires at least $\Omega(N/\sqrt{\text{OUT}})$ time; heavy/light partitioning with Hoeffding intersection estimation gives matching upper bounds. ^{23PODS}

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Fine-grained reductions: direct access with min/max aggregates

In direct access, the database is preprocessed once so that answers to an acyclic free-connex conjunctive query can be enumerated in a user-chosen order with constant delay; the question is which aggregate-driven orders, such as sorting by $\min X$, admit quasilinear-time preprocessing.

Theorem 0.23 — Hard side of the dichotomy. Direct access according to $\min X$ is not possible in quasilinear time if Q has a chordless k -path ($k \geq 3$) between two variables in X , assuming the Hyperclique hypothesis and SETH.

The reduction encodes triangle and hyperclique detection so that a quasilinear-time direct-access structure for $\min X$ would decide hyperclique existence faster than the hypothesis permits. The easy side of the dichotomy is algorithmic: strict total orders are partitioned into partial orders that suitably rearranged join trees enforce, after existential variables are eliminated in linear time. ^{24PODS}

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When to reach for reductions

- **Lower bounds without a from-scratch proof.** Pick the canonical hard problem closest in shape (Set Cover for covering and outlier budgets, InnerProduct for private estimation, hyperclique detection for join patterns, listing for enumeration), build a gadget, and state explicitly which hypothesis the reader inherits.
- **Hardness of approximating, not just deciding.** Make the reduction gap-preserving (an L -reduction), so the source's inapproximability threshold — not merely its NP-hardness — crosses

over to your objective.

- **Fine-grained statements.** When the target is a refined running time (sublinear, output-sensitive, constant delay), condition on a quantitative hypothesis and audit the parameter blow-up of the gadget; a sloppy parameter map makes the bound vacuous in some regime.
- **Reductions also certify tractability.** The same machinery run backwards — encoding your problem into LP feasibility, matchings, or coresets — builds the upper-bound side of a dichotomy.

Induction and Invariants

Induction is the workhorse that never gets top billing. In our tagged SIGMOD’26 corpus, 36 of 228 papers use it (the #3 tool overall); the pairing *correctness* $\times$ *induction*, with 29 papers, is the single most frequent property–tool co-occurrence in the survey, and induction also powers the *upper-bound* pairing in 21 papers. The characteristic shape is not one grand named theorem but thousands of small, unnamed invariants — half-page lemmas saying that one more elimination step, recursive call, or rewrite keeps a component consistent. Induction is thus the instrument of *component* correctness: each cache, rewrite rule, or decomposition step is verified locally, and the global guarantee is assembled from the local ones.

The Tool: Induction on What?

Every inductive argument in the corpus answers two questions: what do we induct on, and what is preserved at each step?

Induction on what? Three substrates dominate.

1. **Execution steps.** An algorithm induces a sequence of states $s_0, s_1, \dots, s_t$, one per operation (a function call, an elimination, a rewrite). One proves a state predicate I — the *invariant* — by induction on i : $I(s_0)$ holds, and each operation preserves I , so I holds in every reachable state and implies the specification at final states. This is the inductive-assertion method of Floyd ²⁵ and Hoare ²⁶, the loop-invariant discipline of algorithms texts ²⁷, and Manna–Pnueli *safety* reasoning: nothing bad ever becomes true ²⁸.
2. **Structural recursion.** The object is built recursively — a tree decomposition, an elimination forest, a recursive pseudo-tree, a hierarchy — and the property is proved on the whole from the property on the parts. The database workhorse is *leaf-peeling*: remove a leaf bag or leaf vertex, apply the induction hypothesis to the smaller object, then extend by one step.

²⁵
²⁶
²⁷
²⁸

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3. **A natural-number measure.** Strong induction on a quantity that strictly decreases: the number of eliminated nodes, the number of bags, or the length of a rewrite sequence; structural induction is routinely converted into this form by choosing a measure ²⁹.

Example 0.24 (Invariant preservation). Given a set of states Σ , an initial state s_0 , and a set of operations Γ , a predicate $I \subseteq \Sigma$ is an *invariant* if (i) $I(s_0)$ holds, and (ii) whenever $I(s)$ holds and s' results from applying an operation of Γ to s , then $I(s')$ holds. By induction on the number of executed operations, every reachable state satisfies I ; any property implied by I at final states is therefore guaranteed.

$$\begin{array}{ccccccc} s_0 & \xrightarrow{op_1} & s_1 & \xrightarrow{op_2} & \dots & \xrightarrow{op_t} & s_t \\ I(s_0) & & I(s_1) & & \dots & & I(s_t) \end{array}$$

Base case $I(s_0)$; every operation preserves I ; hence I holds at every step, and $I(s_t)$ implies the specification.

Structural versus strong induction. Structural induction follows the formation rules of a data structure (P on the immediate substructures yields P on the structure); strong induction on the naturals assumes $P(j)$ for all $j < k$ to derive $P(k)$. The two are interderivable via a measure such as the number of bags of a decomposition; one picks whichever form makes the step case a purely local check.

Exemplars from the Corpus

Induction on elimination steps. Exact resistance-distance computation on graphs of small treewidth eliminates the nodes of a tree decomposition one at a time, updating a pseudo-inverse matrix that must retain a precise block shape ³⁰.

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Lemma 0.25 — Elimination state. Let G be a graph with Laplacian L , and let $U_1 \subseteq U_2 \subseteq V$ be sets of nodes. After the elimination of the nodes of $U_2 \setminus U_1$, the state matrix satisfies

$$S_{|U_2 \setminus U_1|} = \begin{pmatrix} L_{U_1 U_1}^{-1} & 0 \\ 0 & 0 \end{pmatrix}.$$

The induction variable is the number of eliminated nodes, and the induction hypothesis is stated as an explicit invariant of the state: after eliminating k nodes (base case $S_0 = L_{U_2 U_2}^{-1}$), all rows and columns of eliminated nodes are zero, and the block on the surviving nodes is exactly the Schur complement of the remaining nodes ³¹.

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Induction on the run, and on bags. Charting which space–time exponent pairs are feasible for sum–product queries requires algorithms that evaluate a query along a recursive pseudo-tree (RPT) with per-vertex caches ³².

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Lemma 0.26 — Cache invariant. Let $\mathcal{T}$ be a recursive pseudo-tree (RPT) of the sum–product query Q , and let A be an arbitrary vertex of $\mathcal{T}$. Then, during the execution of the algorithm, the function call $\text{solve}(A, \mathbf{x})$ returns the answer to the subquery of Q at A , restricted to the part of the partial assignment $\mathbf{x}$ that is over the variables of A ; furthermore, after the subsequent fillCache call, the cache M_A holds exactly this value.

The induction is over the sequence of function calls of a run: the invariant is that every completed $\text{solve}/\text{fillCache}$ pair leaves M_A storing precisely the restricted subquery answer. The companion plan-dominance proof is structural, proceeding by induction on the number of bags of a tree decomposition: peel a leaf bag, eliminate the variables in that bag but not in its parent’s, and apply the hypothesis to the smaller decomposition ³³.

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Induction on rewrite sequences. A unifying algorithm for hierarchical queries computes full symbolic provenance once and transports it to any target algebra by a homomorphism; its correctness rests on a two-rule elimination procedure on the query itself ³⁴.

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Proposition 0.27 — Elimination procedure for hierarchical queries. A query Q is hierarchical if and only if the following elimination procedure transforms Q into a query with a single nullary atom, i.e. of the form $Q() :- R()$. Repeatedly apply one of the following two rules, until neither rule applies:

- (Rule 1) Eliminate a “private” variable: find a variable $Y \in \text{vars}(Q)$ that occurs in only one atom $R(\mathbf{X})$ in Q , and replace $R(\mathbf{X})$ with $R'(\mathbf{X} \setminus \{Y\})$, where R' is a new relation symbol.
- (Rule 2) Eliminate a “duplicate atom”: find two distinct atoms $R_1(\mathbf{X})$, $R_2(\mathbf{X})$ over the same variables $\mathbf{X}$ and replace the two by a single atom.

The induction variable is the length of the rewrite sequence, and the invariant holds in both directions: a hierarchical query not yet of the form $Q() :- R()$ always admits Rule 1 or Rule 2 and stays hierarchical, while either rule applied to a non-hierarchical query leaves it non-hierarchical. The same stepwise induction carries the provenance theorem — after each rule application, the output is the φ -image of the symbolically annotated one ³⁵.

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Bottom-up induction over the query tree. Do regular-path atoms make acyclic conjunctive queries harder to evaluate than relational ones? For combinatorial algorithms the answer is no, and the guarantee is quantitative ^{36PODS}.

Theorem 0.28 — Acyclic CRPQ evaluation. Let Q be any acyclic CRPQ, and G be an input edge-labeled graph of size N . Then Q over G can be answered in time

$$O(N + N \cdot \text{OUT}^{1-1/\max(\text{fn-fhtw}(Q), 2)} + \text{OUT})$$

in data complexity, where OUT is the output size and $\text{fn-fhtw}(Q)$ is the free-connex fractional hypertree width of Q .

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Correctness and runtime are proved together by bottom-up induction over the acyclic structure of Q (reduced first to bound-connected components and free-leaf CRPQs): the invariant at each subtree is that the partial evaluation materializes exactly the answers of the subquery rooted there, with heavy/light splitting keeping every intermediate within the budget $\Delta = \text{OUT}^{1-1/\ell}$ ^{37PODS}.

When to reach for induction

- A component has state evolving by local steps and the claim is “nothing drifts” — caches, elimination pipelines, rewrite rules. The property is a safety property: prove an invariant over the state, not a formula about the final value.
- Choose the induction variable first: execution steps for imperative components; the recursive structure (tree decomposition, pseudo-tree, hierarchy) for recursive ones. Leaf-peeling is the default move on trees.
- Make the invariant strong enough to imply the specification. If the step case will not close, strengthen it (add the block shape, the cache contents, hierarchality) — strengthening is the normal iteration of the method, not a failure.
- Pair the invariant with a strictly decreasing measure (bags, nodes, or rule applications left) to obtain termination and runtime bounds — this is how induction also powers the *upper-bound* pairing (21 papers).
- Audit the step case over *every* operation, and check the trivial base cases (empty input, single bag, zero eliminations): one missed case silently breaks the entire argument.

Estimation Theory

Estimators: the interface between sampling and guarantees

Database systems rarely compute quantities exactly: sketches compress frequency vectors, samplers select a few rows or a few random spanning trees, and privacy mechanisms deliberately add noise. Estimation theory is what lets a paper say something rigorous about the resulting number. Whenever a data structure or mechanism reports a value $\hat{\theta}$ together with a claim of the form “within ε of the truth with probability at least $1 - \delta$,” an *estimator* has been constructed and its error has been bounded. The three classical questions transfer verbatim: is $\hat{\theta}$ *unbiased*, how large are its *variance* or mean-squared error, and can *any* estimator in the same situation do better, i.e., what does a *minimax* or information-theoretic lower bound permit ³⁸? In the SIGMOD’26 corpus, estimation theory is used by 27 of the 228 tagged papers and is the most frequent companion of the *error-bound* (16 papers) and *probabilistic-guarantee* (13 papers) tags: it is the tool that turns a data reduction into a guarantee.

³⁸; and

The workflow typically runs backwards from the target guarantee: build a sketch or sampler; read an estimator off it; bound the estimator’s variance; convert variance into a confidence statement with a concentration inequality; pay a union bound over the keys, nodes, or rounds that must be simultaneously correct; and, in the strongest papers, prove a matching lower bound showing the error or the space is optimal.

The tool

Bias–variance decomposition. For an estimator $\hat{\theta}$ of a parameter θ ,

$$\text{MSE}_{\theta}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \underbrace{(\mathbb{E}[\hat{\theta}] - \theta)^2}_{\text{squared bias}} + \underbrace{\text{Var}(\hat{\theta})}_{\text{variance}}, \quad (1)$$

and every design decision in a sketch or mechanism is a move between the two terms. Unbiasedness ($\mathbb{E}[\hat{\theta}] = \theta$) is prized because variances *add* over independent contributions — per-key counters, per-round messages, per-node estimates — so one variance computation feeds

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Chebyshev or Chernoff bounds and yields the (ε, δ) statement directly³⁹. Its fragility matters equally: unbiasedness is destroyed by nonlinear post-processing, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$, so estimating x^p from an unbiased estimate of x requires a Taylor-series or geometric-mean correction rather than plugging in.

Unbiasedness versus minimax. A single unbiased estimator says nothing about optimality. The minimax risk

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}[(\hat{\theta} - \theta)^2]$$

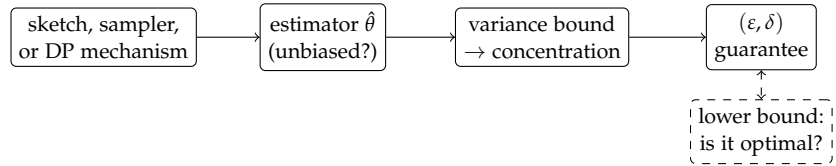
40

benchmarks *all* estimators at once. For unbiased estimators the Cramér–Rao inequality $\text{Var}_{\theta}(\hat{\theta}) \geq 1/I(\theta)$, with $I(\theta) = \mathbb{E}[(\partial_{\theta} \log p_{\theta}(X))^2]$ the Fisher information, gives a local benchmark; two-point (Le Cam) and Fano arguments give distribution-free ones⁴⁰. In DB papers the “information” is usually combinatorial rather than Fisherian: the sensitivity of a query under neighboring inputs, the norm of the stream, or the entropy of any encoding of the sketch. The last turns estimation error into a space lower bound: if the estimator must be (E, δ) -accurate, the sketch state must be incompressible below the entropy of what it has to distinguish⁴¹.

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The estimator-to-bound pipeline. Corpus papers instantiate the same five-step pipeline: (i) construct $\hat{\theta}$ from a sketch, a sample, or a randomized mechanism, and check its expectation; (ii) bound $\text{Var}(\hat{\theta})$ or a sub-Gaussian proxy in terms of structural parameters (stream norms, degrees, sensitivities); (iii) apply a concentration inequality to pass from variance to a per-quantity (ε, δ) statement; (iv) union-bound over all coordinates that must hold simultaneously, paying a $\log(\#\text{quantities})$ factor; (v) match with a lower bound (variance, Le Cam/Fano, or entropy) to certify optimality⁴².

42 ; ; and



The estimator pipeline: every exemplar below walks left to right; the strongest also close the loop.

Exemplars from the SIGMOD’26 corpus

Perfect ℓ_p sampling from exponential variables (streaming)

One-pass streaming algorithms must output a single index i distributed with probability proportional to $|f_j|^p$, the ideal ℓ_p sample; the classical

difficulty is that the stream must be tracked under this unusual distribution. The paper treats the output index itself as the estimator and bounds its accuracy per coordinate.

Theorem 0.29 — Theorem 1 of ⁴³: ℓ_p **sampler.** For any constant $p \in (0, 2]$, there is a one-pass streaming algorithm that runs in space $\tilde{O}(\log^{c(p)} n \cdot \log(1/\delta))$ and outputs an index i such that, with probability at least $1 - \delta$, for every $j \in [n]$,

$$\Pr[i = j] = \frac{|f_j|^p}{\|f\|_p^p} \pm \frac{1}{\text{poly}(n)}, \quad c(p) = \begin{cases} 1 & 0 < p < 2, \\ 2 & p = 2. \end{cases}$$

Corollary 0.30 — Corollary 3.3 of ⁴⁴. Let $f \in \mathbb{R}^n$ be a frequency vector and let $(e_1, \dots, e_n)$ be i.i.d. exponential random variables with rate 1. Define $z_i = f_i / e_i^{1/p}$ and let $(D^{(1)}, \dots, D^{(n)})$ be the anti-rank vector of the vector $(z_1^{-p}, \dots, z_n^{-p})$. Then

$$\Pr[D^{(1)} = i] = \frac{|f_i|^p}{\|f\|_p^p}.$$

What is estimated is a whole distribution rather than a scalar, and the error bound is an ℓ_∞ bound on the sampled probabilities. The estimator trick is the exponential race: by scaling, $e_i / f_i^p = z_i^{-p}$ is exponential with rate $|f_i|^p$, so the minimum is attained at i with exactly ℓ_p probability; the streaming algorithm then only has to track the rescaled vector z (via BPTree, CountSketch, and AMS sketches), repairing the nonlinear x^p step with Taylor-series and geometric-mean estimators, with Chebyshev and union bounds supplying the $\pm 1/\text{poly}(n)$ accuracy. ⁴⁵PODS

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Unbiased Laplace-noised walk counting (local differential privacy on graphs)

Counting k -line walks in a graph where every node privatizes its own view (edge-local DP) forces fresh Laplace noise at each of k synchronous rounds, so the naive error compounds with k ; the question is how much accuracy a k -round message-passing mechanism can retain.

Theorem 0.31 — Theorem 4.2 of ⁴⁶. For any graph G , walk length k , and parameters ϵ and β , the k -round mechanism $\mathcal{M}_{k\text{-wk}}$ satisfies $\mathbb{E}[\mathcal{M}_{k\text{-wk}}(G, \epsilon)] = W_k$, and with probability at least $1 - \beta$,

$$|\mathcal{M}_{k\text{-wk}}(G, \epsilon) - W_k| \leq k\gamma\sqrt{N} (d(G) + \gamma)^{k-1} = \tilde{O}(\sqrt{N d(G)^{k-1}}),$$

where $\gamma = 2k\sqrt{8\log(2kN/\beta)}/\epsilon = \tilde{O}(1)$, with N the number of nodes and $d(G)$ a degree parameter of G .

What is estimated is W_k , the number of k -line walks; the bound is unbiasedness plus a high-probability additive error of $\tilde{O}(\sqrt{N d(G)^{k-1}})$,

improving on the typical $O(N^{k/2})$ of naive Laplace counting. The estimator trick is sensitivity control by induction: each round adds per-node Laplace noise scaled by the running maximum $\|X^{(\ell-1)}\|_\infty$, a Laplace concentration bound (failure budget β'/N per node per round) yields $\|X^{(\ell)}\|_\infty \leq (d(G) + \gamma)^\ell$, and union bounds close the loop; a complementary variance lower bound of $\Omega(N^{k+1})$ for the baseline estimator, obtained by summing variances over all oriented k -line walks, shows the multi-round design is genuinely needed. ⁴⁷

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Variance recursion and an entropy lower bound for expandable sketches

A frequency-estimation (FE) sketch for streams of unknown, unbounded length must expand as the stream grows; the paper asks both how accurate such a sketch can be and how much memory the expansion forces, matching the two sides.

Lemma 0.32 — Variance of the sketch ⁴⁸. Like CountSketch, the sketch provides unbiased estimates. Moreover, letting $V(F)$ denote its estimation variance when using a size function $W(\cdot)$ and one array on a stream with squared ℓ_2 -norm $F = \sum_y (f(y))^2$, we have at the end of epoch i that

$$V(F_{i+1}) \leq \frac{1}{W_i} \cdot \sum_{j=0}^i 2^{i-j} \cdot (F_{j+1} - F_j),$$

where F_j denotes the squared ℓ_2 -norm of the stream at the start of epoch j .

Theorem 0.33 — Space lower bound ⁴⁹. Consider an FE sketch with only overestimation errors that processes growing streams with an error bound of $E(N)$ and a confidence of $1 - \delta$. Such a sketch must use at least

$$\frac{N}{E(N)} \cdot \left[\Omega\left(\log \frac{1}{\delta} + \log \log \frac{N}{E(N)}\right) - \Theta(1) \right]$$

bits of memory at some point in time while processing a stream.

What is estimated are per-key frequencies, unbiasedly as in CountSketch; the error bound is a variance recursion over geometrically spaced epochs, each past expansion contributing with weight 2^{i-j} , and Markov/Chebyshev convert it into the stated confidence. The complementary trick is the lower bound: an entropy-incompressibility argument (by reduction to filters) shows any sketch achieving error $E(N)$ at confidence $1 - \delta$ must pay the displayed bits, and the constructed sketch lands within a small logarithmic factor of it — the estimation-theoretic ideal of a matched pair. ⁵⁰

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Concentration-chosen sample sizes for influence minimization

Blocking nodes to curb time-critical misinformation spread requires maximizing a benefit function $D(\cdot)$ that is $\#P$ -hard to evaluate and non-submodular, so greedy must run on sampled surrogates; the question is how many samples suffice for the greedy choice to provably transfer to the true objective.

Theorem 0.34 — Theorem 6.7 of ⁵¹. Given $0 \leq \varepsilon, \delta \leq 1$, CBFM returns B^* satisfying

$$\Pr [D(B^*) \geq \tau(1 - \frac{1}{e} - \varepsilon) \cdot D(B_0)] \geq 1 - 3\delta,$$

where B_0 denotes an optimal solution.

What is estimated is $D(B)$ itself, by an unbiased estimator $\hat{D}(B)$ built from reverse-score sample sets (GRS/TRS) whose coverage terms are unbiased; the error bound is an end-to-end $(1 - 1/e - \varepsilon)$ -style approximation holding with probability $1 - 3\delta$, where τ reflects that greedy optimizes D through the submodular surrogates sandwiching it ($D_L \leq D \leq D_U$). The estimator trick is to derive the sample threshold analytically: concentration bounds fix the maximal sample size $\theta_{\max}$, and three union-bound events account for the 3δ . ⁵²

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How many random trees? (biharmonic distance)

Pairwise biharmonic distance requires quantities from the square of the Laplacian pseudoinverse, which no near-linear graph algorithm computes exactly; the paper reformulates the distance and estimates each term by averaging over randomly sampled spanning trees.

Theorem 0.35 — Theorem 6.3 of ⁵³. If $\omega_t = d_v^2 \Delta^2 \log(2n/\delta)/(2\varepsilon^2)$, FastTree samples ω_t trees, and the corresponding estimators satisfy that for each node u ,

$$|x_a(u) - \tilde{x}_a(u)| \leq \varepsilon \quad \text{and} \quad |x_b(u) - \tilde{x}_b(u)| \leq \varepsilon.$$

What is estimated are the two terms x_a, x_b of the reformulated distance (a pivoted identity rewrites $\beta(s, t)$ via $\hat{L}_v^{-1}$ and a rank-structured correction), each by the average of ω_t unbiased tree-sampling estimators grounded in electrical-network/Kirchhoff identities; the error bound is per-node ε -accuracy from Chernoff–Hoeffding plus a union bound over all n nodes. The trick is visible in the sample count itself: it is exactly the canonical Hoeffding size (range scale $d_v \Delta$, with d_v the pivot degree and Δ a graph parameter) squared, times $\log(2n/\delta)$, over $2\varepsilon^2$ — the matrix inverse is replaced by an expectation over absorbing random walks. ⁵⁴

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When to reach for estimation theory

- You report a number derived from a sketch, a sample, or a randomized mechanism, and must attach error bars or an (ε, δ) confidence to it.
- Unbiasedness is the workhorse: it makes variances add across keys, rounds, or nodes, so one variance bound plus Chebyshev/Chernoff delivers the guarantee.
- Nonlinearity breaks unbiasedness ($\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$); fix it with Taylor-series or geometric-mean corrections instead of ignoring the bias.
- To size a structure, invert the concentration bound: sample counts of the form $\omega \approx \text{range}^2 \log(n/\delta)/(2\varepsilon^2)$ should fall out analytically, not be tuned.
- To certify optimality, pair the upper bound with a variance, Cramér–Rao, Le Cam/Fano, or entropy-incompressibility lower bound; matching within logarithmic factors is the gold standard.
- Remember the two hidden costs: union bounds charge $\log(\#\text{quantities})$, and minimax bounds rate the estimator class, not your implementation.

Adversarial Constructions: How Lower Bounds Get Built

An upper bound is a promise; an impossibility result is the proof that no better promise exists. The second kind of theorem is not derived from the first — it is *built*. To show that no algorithm, language, or sketch in a model can do better, the author must construct the world in which every allowable procedure fails: a pair of instances, a hard distribution, a stream of updates engineered so that succeeding is impossible and failing is provable. That engineered world is the *adversary*, the workhorse of impossibility. In our corpus the tool is used by 21 of 228 tagged papers, and it owns the survey’s signature pairing: 13 of those papers prove an upper bound *and* its matching adversarial lower bound in the same article. The pairing is no accident — the upper bound tells you what the adversary must force; the adversary certifies that the upper bound is the end of the line.

The Tool: Anatomy of an Adversarial Argument

Strip away the domain vocabulary and adversarial arguments share a three-part skeleton.

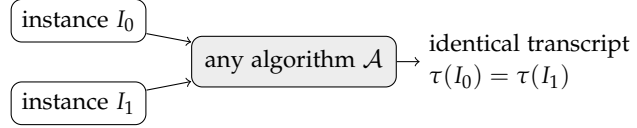
The world: an instance family. Fix the model of computation or observation — the rows of a linear sketch, a stream of tuple inserts and deletes, the syntax of an algebra, a class of colored graphs. Inside that model, design the instance family: often two planted instances I_0 and I_1 , or, via Yao’s minimax principle ⁵⁵, a single hard input distribution against which the best deterministic procedure is analyzed.

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Forced behavior or information blindness. Argue that *every* procedure in the model is constrained on the family, for one of two reasons. *Information blindness*: the observation available to the algorithm (a sketch, a transcript, a vector of walk counts) simply does not contain the bit that distinguishes the instances. *Forced behavior*: the model’s own syntax caps a resource — e.g. an expression of size ℓ can multiply tuple

multiplicities by at most 2^ℓ — and the adversary exhibits an instance whose correct answer exceeds every cap.

The two-instance indistinguishability pattern. The decisive move: instances I_0 and I_1 produce identical observations yet demand different answers, so any output is wrong on one of them.



yet the correct answers differ: $f(I_0) \neq f(I_1)$,
so $\mathcal{A}$ must fail on one of them

Imported adversaries: reductions and communication complexity. When no direct planting suffices, the adversary is imported. A conditional lower bound is an adversary-as-reduction: an effective procedure converts instances of a believed-hard problem (k -clique, OuMv, SETH) into instances or update sequences of yours, so any fast algorithm would refute the conjecture — the dominant pattern for dynamic graph and query problems⁵⁶. Communication complexity hands out prefabricated two-player adversaries — index, set-disjointness, gap-Hamming — whose transcripts are information-blind by construction⁵⁷; sketching lower bounds in databases have ridden this route since the frequency-moment bound⁵⁸. And when the object under attack is a *language* rather than a cost, the adversary is a witness object, from Moon–Moser’s triangle graph⁵⁹ to the counterexamples that have measured relational completeness since Codd⁶⁰.

Exemplars from the Corpus

Walk-Equivalent Graph Pairs and GNN Expressiveness

MPLang is a matrix-programming language capturing GNN-style computation over embedded graphs; its affine fragment A-MPLang omits activation functions. What can the affine fragment even see?

Corollary 0.36 The following are equivalent for pointed d -embedded graphs (G_1, γ_1, v_1) and (G_2, γ_2, v_2) :

1. $e(G_1, \gamma_1)(v_1) = e(G_2, \gamma_2)(v_2)$ for every A-MPLang expression e of A^+ -depth n over d -embeddings;
2. $((A^+)^i 1)(G_1, \gamma_1)(v_1) = ((A^+)^i 1)(G_2, \gamma_2)(v_2)$ and $((A^+)^i P_j)(G_1, \gamma_1)(v_1) = ((A^+)^i P_j)(G_2, \gamma_2)(v_2)$ for all $i \in [n]$ and $j \in [d]$.

The condition in the second item defines (G_1, v_1, γ_1) and (G_2, v_2, γ_2) being n -walk equivalent.

A normal-form theorem shows every affine expression of depth n is a fixed polynomial in the walk-count features of item 2, so any two graphs agreeing on those counts are indistinguishable to the entire fragment — the textbook two-instance pattern, with walk statistics as the shared transcript. A single walk-equivalent pair that disagrees on the target query kills the fragment, and red–blue symmetric trees (with induction on expression shape) extend the witnessing to separate activation-function classes: any Σ -MPLang expression must stay symmetric exactly where a ReLU threshold can break symmetry. ^{61PODS}

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Sketching Dimension for ℓ_2 Sampling

A linear ℓ_2 -sampling sketch observes Sv and must output coordinate i with probability roughly $v_i^2 / \|v\|_2^2$. How many rows must S have, and how many bits store Sv ?

Theorem 0.37 An (ϵ, δ) - ℓ_2 -sampler with frequency estimation requires a sketching dimension of at least $\Omega(\frac{1}{\epsilon} \log^2 n + \frac{2}{\epsilon} \log n)$. Moreover, storing such a sketch for vectors with $\text{poly}(n)$ -sized entries requires $\Omega((\log^3 n + \frac{2}{\epsilon} \log^2 n) / \epsilon)$ bits.

The adversary is a planted distribution made rigorous by Yao’s principle: m short Gaussian spikes sit on isotropic background noise, so each row of the sketch observes a location mixture $X \sim N(v_I, I_d)$ with I uniform and $\|v_i\| \leq 10 \ln m$. The paper’s Lemma 2 bounds recovery of I from X by $2m^{-1/2}$ — the spikes are information-blind to any decoder — and a payout-game argument converts failed identification into sampling error. ^{62PODS}

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Conditional Lower Bounds for Dynamic Query Maintenance

A dynamic algorithm maintains the answers to a fixed conjunctive query Q under single-tuple inserts and deletes, with amortized update time and enumeration delay. Which maintenance times are achievable without blowing up the update cost?

Theorem 0.38 Assuming the combinatorial k -clique conjecture, no dynamic combinatorial algorithm can maintain Q with amortized update time $O(|D|^{(k-1)/k-\epsilon})$ and support constant-delay enumeration, for any constant $\epsilon > 0$.

Here the adversary is imported: a reduction encodes k -cycle detection (via color-coding hash families) and OuMv tensor products into an update sequence on the database, so a faster maintenance algorithm

would solve the conjecture-hard problem. The reduction *is* the adversary — it generates the stream adaptively, and indistinguishability is inherited from the source problem rather than planted directly. ⁶³PODS

Expressiveness: Relational Algebra Without Division

On $\mathbb{K}$ -databases every tuple carries a semiring value, and the paper asks which fragments of calculus and algebra still coincide in the spirit of Codd’s theorem ⁶⁴. One operation resists every expression.

Theorem 0.39 There is no expression of basic relational algebra BRA that defines the division operation $\div$ on $\mathbb{N}$ -databases.

This is forced behavior by a counting ceiling: structural induction shows every BRA expression E on inputs of size n has support at most $n^{\ell(E)}$ and multiplicities at most $p_E(n) \cdot 2^{\ell(E)}$ — single-exponential growth in expression size. The adversary is one 2^n -multiplicity division instance, which outruns the budget of every fixed expression; no pair of instances is needed, only growth rates. ⁶⁵

Output-Sensitive Witness Graphs for Defective Cliques

A k -defective clique is a vertex set missing at most k edges; enumeration algorithms are judged by search space relative to output. How large must the output itself be in the worst case?

Theorem 0.40 The worst-case output size of maximal k -defective clique enumeration is $\Omega(3^{n/3} \cdot n^k)$ when k is a constant.

The adversary is a Moon–Moser-style witness graph ⁶⁶: disjoint triangles already force an independent three-way choice per block ($3^{n/3}$ maximal cliques), and each of the k units of permitted defect contributes another polynomial factor of choices. The construction bounds *every* enumeration algorithm — the output itself is that large — and thereby certifies the paper’s worst-case-optimal search space as tight. ⁶⁷

When to Build an Adversary

- You have (or plan) an upper bound and need to certify optimality: aim the adversary at exactly the resource your algorithm touches — this is the 13-paper pairing, not a coincidence.
- The question is “can *any* method in this model do better” or “is this language expressive enough”, not “is my implementation correct”.
- Recipe: fix the observation model; find what all procedures

must share (a transcript, a growth ceiling, a symmetry); plant two instances that share it but demand different answers; or import the pair from a conjecture-hard problem via a reduction, or from communication complexity.

- Against randomized algorithms, invoke Yao's principle first and design a hard *distribution*; the analysis then runs against deterministic procedures.
- If every two-instance attempt fails, take the hint: perhaps the model is sufficient. Try the matching upper bound before fighting the lower bound.

Differential Privacy Composition

Among the 26 SIGMOD'26 papers tagged `privacy-dp`, composition accounting is close to a monoculture: 69% of them assemble their end-to-end privacy guarantee by composing mechanisms, and the tool `dp-composition` is used by 18 of the 228 tagged papers overall, its strongest pairing being `privacy-guarantee` $\times$ `dp-composition` (16 papers). The arithmetic itself is shifting: the exemplars below still mostly add plain (ϵ, δ) budgets, but the frontier results state composition for Rényi DP and f -DP, where the composed object is a whole divergence curve rather than a pair of scalars.

The tool

Definition 0.41 (Differential privacy). A randomized mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ satisfies (ϵ, δ) -differential privacy if for all neighboring datasets $D \sim D'$ (differing in one record, edge, or node) and all measurable $S \subseteq \mathcal{R}$,

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \cdot \Pr[\mathcal{M}(D') \in S] + \delta.$$

With $\delta = 0$ the guarantee is *pure* ϵ -DP [?].

Five primitives, one line each:

- **Sequential composition.** An adaptive sequence of releases with budgets (ϵ_i, δ_i) satisfies $(\sum_i \epsilon_i, \sum_i \delta_i)$ -DP: budgets add [? ?].
- **Post-processing immunity.** Any function of an (ϵ, δ) -DP output that does not touch the raw data is itself (ϵ, δ) -DP: analysis after the release is free [?].
- **Group privacy.** Datasets at distance λ inherit $(\lambda\epsilon, \lambda\delta)$ -DP: the guarantee degrades linearly in group size [?].
- **The Rényi ladder.** The Gaussian mechanism (noise $\mathcal{N}(0, \sigma^2 I)$, ℓ_2 -sensitivity Δ_2) is $(\alpha, \alpha\Delta_2^2/(2\sigma^2))$ -RDP for every $\alpha > 1$; RDP composes by adding the $\rho(\alpha)$; and (α, ρ) -RDP converts to $(\rho(\alpha) + \log(1/\delta)/(\alpha - 1), \delta)$ -DP, optimized over α — the engine behind $O(\epsilon\sqrt{k})$ rather than $k\epsilon$ for k releases [? ?]. f -DP makes the divergence curve itself the composed object [?].

- **Amplification by subsampling.** An ϵ -DP mechanism run on a p -rate Poisson subsample satisfies $\log(1 + p(e^\epsilon - 1))$ -DP: smaller samples buy smaller budgets [?].

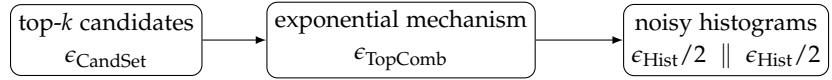
Exemplars from SIGMOD'26

Budget ledgers for private cluster explanations. The paper privately explains a given clustering by selecting the attribute combination that best accounts for it: top- k candidate selection, then an exponential mechanism over combinations, then per-cluster histograms ⁶⁸.

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Proposition 0.42 — Sequential composition. Let $\mathcal{M}_1 : \mathcal{D} \rightarrow \mathcal{R}_1$ and $\mathcal{M}_2 : \mathcal{D} \times \mathcal{R}_1 \rightarrow \mathcal{R}_2$. Suppose $\mathcal{M}_1$ satisfies ϵ_1 -DP and, for every $y \in \mathcal{R}_1$, $\mathcal{M}_2(\cdot, y)$ satisfies ϵ_2 -DP (as a function of its first input). Define $\mathcal{M}_3 : \mathcal{D} \rightarrow \mathcal{R}_2$ by $\mathcal{M}_3(D) = \mathcal{M}_2(D, \mathcal{M}_1(D))$. Then $\mathcal{M}_3$ satisfies $(\epsilon_1 + \epsilon_2)$ -DP.

The end-to-end guarantee is the budget sum $(\epsilon_{\text{CandSet}} + \epsilon_{\text{TopComb}} + \epsilon_{\text{Hist}})$ -DP, while inside the histogram stage the clusters are disjoint, so per-attribute and per-cluster releases compose *in parallel*, each contributing $\epsilon_{\text{Hist}}/2$ — pure multiplicative e^ϵ arithmetic, no divergence needed.



sequential total: $\epsilon_{\text{CandSet}} + \epsilon_{\text{TopComb}} + \epsilon_{\text{Hist}}$

Node-DP from edge-DP via group privacy. N2E converts any edge-DP graph mechanism into a node-DP one by clipping the input graph to a degree bound Δ^* that is itself privately approximated, then running the edge mechanism on the clipped graph ⁶⁹.

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Lemma 0.43 — Group privacy. If a mechanism $\mathcal{M}$ satisfies (ϵ, δ) -differential privacy, then for any two graphs G, G' with $d(G, G') = \lambda$ (i.e., differing by λ edges or nodes) and any subset of outputs $S \subseteq \mathcal{R}$,

$$\Pr[\mathcal{M}(G) \in S] \leq e^{\lambda\epsilon} \cdot \Pr[\mathcal{M}(G') \in S] + \lambda\delta.$$

The mechanism satisfies $(\lambda\epsilon, \lambda\delta)$ -DP for the group of λ edges or nodes.

The composition primitive doing the lifting is group privacy: a one-node change is a group of λ edge changes, so the edge-DP calculus transfers to node-DP neighborhoods. The degree search spends sparse-vector-technique queries at a logarithmic scale, so only logarithmically many sequential compositions are consumed before post-processing turns the noisy degree into the clipping Δ^* ; the whole framework satisfies (ϵ, δ) -node-DP ⁷⁰.

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Per-record budgets: composition of functions, not scalars. Per-record DP replaces the global ϵ by a budget function $\epsilon_{\mathcal{E}}(r)$ (a bank may assign wealthier clients smaller budgets), and estimates sums/counts by partitioning records into geometrically doubling budget slices, one Laplace mechanism per slice ⁷¹.

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Lemma 0.44 — PrDP parallel composition. Let $R_1, \dots, R_k$ be disjoint subdomains and $\mathcal{M}_1, \dots, \mathcal{M}_k$ a series of $\mathcal{E}_i$ -PrDP mechanisms. Their parallel composition $\mathcal{M}(D) := (\mathcal{M}_1(D_{R_1}), \dots, \mathcal{M}_k(D_{R_k}))$ satisfies $\mathcal{E}'$ -PrDP, where $\mathcal{E}'(r) = \max\{\mathcal{E}_i(r)\}_{i=1}^k$ for all $r \in [U]^d$.

Here the composed object is a budget *function*: releases on disjoint subdomains take a pointwise maximum instead of a sum, so a record is only ever charged the largest budget of any slice touching it. Near-optimality of the resulting error is certified against down-neighborhood lower bounds ⁷².

72

Concurrent composition for continual mechanisms. Continual mechanisms answer streams of queries; the hard question is what composition survives an adversary who adaptively and *concurrently* interleaves access to many such mechanisms ⁷³.

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Theorem 0.45 — Concurrent composition of continual mechanisms, informal. The concurrent composition of continual mechanisms $\mathcal{M}_i$ that are (ϵ_i, δ_i) -DP with respect to a verification function f_i is (ϵ, δ) -DP for the same (ϵ, δ) that holds for composing noninteractive mechanisms that are (ϵ_i, δ_i) -DP. This allows the adversary to adaptively and concurrently choose the neighboring sequences fed to each $\mathcal{M}_i$ based on the responses received from the other mechanisms.

What carries the guarantee is a simulation: every (ϵ, δ) -DP interactive mechanism can be realized by interactive post-processing of randomized-response outputs, so the divergence arithmetic of ordinary composition transfers verbatim to concurrent adversaries. The same paper pushes the frontier from (ϵ, δ) to f -DP and Rényi DP, complete with a counterexample showing where intuition breaks — the direction in which the currency is moving. ⁷⁴PODS

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Composing obliviousness in a multi-way join. The differentially oblivious join evaluates queries in untrusted memory while leaking only the input size $|I|$ plus a DP-noisy upper bound $\widehat{OUT}$ on the join size ⁷⁵.

75

Theorem 0.46 — Meta-join. Algorithm 1 is $(\epsilon_1 + \epsilon_2, \delta_1 + \delta_2)$ -DO.

Two primitives carry it: an (ϵ_1, δ_1) -DO subroutine releases the (ϵ_2, δ_2) -DP bound $\widehat{OUT}$, and the advised join's access pattern is a deterministic function of $|I|$ and $\widehat{OUT}$ only. Sequential composition

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adds both coordinates, and post-processing immunity is precisely what makes an *access pattern* a legitimate object to compose ⁷⁶.

When to reach for DP composition

Reach for it whenever a system releases *several* interacting statistics and the privacy proof is a ledger: candidate selections, noisy summaries, clipped releases. Then budgets add (exemplars 1 and 5 above, with δ 's summed too in the approximate case), disjoint data slices switch the arithmetic from sums to maxima (exemplars 1 and 3), and group privacy reduces an expensive neighborhood (node-DP) to a cheap one (edge-DP) at linear cost in group size (exemplar 2). Reach for divergence arithmetic — RDP or f -DP curves — when many releases would make summed ϵ 's vacuous, buying $O(\epsilon\sqrt{k})$ over $k\epsilon$ (the frontier of exemplar 4); reach for sub-sampling amplification when releases are computed on samples anyway. Post-processing immunity ships every downstream artifact — clipped graphs, access patterns, indices — for free. Skip the machinery when a single calibrated release suffices, or when every output is a post-processing of one DP output: composing there only overcounts.

Exchange and Greedy Arguments

Few tools rose as sharply through our SIGMOD’26 corpus as this one: exchange/greedy arguments tagged 4 papers in the arXiv-visible cohort and jumped to 17 once the ACM-only mass arrived—one of the largest risers in the survey. The reason is a uniform paper shape: prove your selection problem NP-hard by reduction, then buy a polynomial-time algorithm back with a provable approximation factor. In our tag co-occurrences the tool pairs with hardness 10 times and with an approximation-ratio property 9 times—the signature of “NP-hard, but structured.” When the structure is submodularity, a covering constraint, or a metric, the greedy (or local-search) algorithm comes with a ready-made certificate, and the paper’s theory section shrinks to checking the hypothesis.

The Tool: Exchange Arguments and Their Guarantees

An exchange argument certifies a greedy choice by showing that any solution avoiding it can be *transformed* into one that makes it, at no loss. In a matroid, independent sets admit pairwise exchanges (if S is independent and smaller than independent S' , some $x \in S' \setminus S$ keeps $S \cup \{x\}$ independent), so greedy is *exact* for any linear objective ⁷⁷. Outside matroids exactness dies, but three structures keep a quantitative exchange certificate alive.

Definition 0.47 (submodularity). $f : 2^U \rightarrow \mathbb{R}_{\geq 0}$ is *submodular* if $f(S \cup \{x\}) - f(S) \geq f(T \cup \{x\}) - f(T)$ whenever $S \subseteq T$ and $x \in U \setminus T$; it is *monotone* if $S \subseteq T \Rightarrow f(S) \leq f(T)$.

- **Monotone submodular, cardinality k .** Diminishing returns let greedy’s next pick match the average marginal value of the optimum’s remaining elements; k geometric decay steps compound to $f(S_{\text{gre}}) \geq (1 - 1/e) \cdot \max_{|B|=k} f(B)$, and $1 - 1/e$ is tight unless $P = NP$ ⁷⁸.
- **Submodular without monotonicity.** Randomized greedy variants retain a $1/2$ guarantee (exactly, via double greedy in the unconstrained case) and degrade gracefully under cardinality constraints ⁷⁹.

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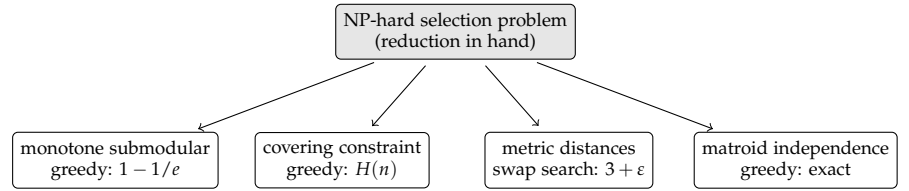
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80

- **Covering constraints (set cover).** To cover a universe at minimum weight, charge each newly covered element the amortized cost of the set covering it; the averaging step against an optimal cover yields the harmonic factor $H(n) = \sum_{i=1}^n 1/i$ ⁸⁰.

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- **Metric structure.** With triangle-inequality distances, exchanging one center for another (local search) reassigns points through two triangle inequalities: single swaps give $(3 + \epsilon)$ for k -median, multi-swaps $(2 + \epsilon)$ ⁸¹.



Which structure earns which guarantee; non-monotone objectives fall to randomized variants at $1/2$.

Five Views from the Corpus

Monotone Submodular Diversity: Greedy at $1 - 1/e$

Selecting the k most *diverse* answers to a fixed conjunctive query, where diversity is defined through a volume assignment Ω over a universe U of possible solutions.

Theorem 0.48 — volume-based diversity. Let Ω be any volume assignment over a universe U of possible solutions. Then δ_Ω is always monotone and submodular. Since Q is fixed and $Q(D)$ is computable in polynomial time over D , the greedy strategy that starts from $S = \emptyset$ and in each iteration adds the tuple $t \in Q(D)$ maximizing the marginal diversity $\delta_\Omega(S \cup \{t\})$ returns a k -subset S with

$$\delta_\Omega(S) \geq \left(1 - \frac{1}{e}\right) \cdot \max_{B \subseteq Q(D), |B|=k} \delta_\Omega(B).$$

The factor is earned by pure Nemhauser–Wolsey–Fisher structure: a measure-algebra argument shows every volume assignment makes δ_Ω monotone and submodular, and greedy inherits $1 - 1/e$ —which is tight (maximum coverage is hard beyond $1 - 1/e$ unless $P = NP$). The contrast is instructive: for the paper’s distance-based analogues under Hamming or Euclidean metrics the best known ratios are strictly worse and degrade with ϵ or k , while every polynomial-time-computable volume assignment gets the clean factor ⁸².

82

NP-Hard Representative Sampling: Lazy Greedy at $1 - 1/e$

Choosing a representative subset of large-scale vector data under a budget: the paper first proves its Problem 1 (Representative Sampling) NP-hard, then turns the budgeted objective into submodular form.

Lemma 0.49 — submodularity of the sampling objective. Let F be the sampling objective, built from per-point gains g_s that are each nondecreasing and concave. Then $F : 2^U \rightarrow \mathbb{R}_{\geq 0}$ is nondecreasing and submodular.

Theorem 0.50 — greedy guarantee. Let B be the budget and let A_{gre} be the size- B output of (lazy-)greedy. Then

$$F(A_{\text{gre}}) \geq \left(1 - \frac{1}{e}\right) \max_{|A| \leq B} F(A).$$

Concavity of every per-point gain g_s is the whole proof: sums of concave lifts of coverage are submodular, and the classical bound then prices the greedy output at $1 - 1/e$ —a one-page certificate for a problem the reduction just declared hopeless exactly ⁸³.

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Greedy Risk Coverage: A Set-Cover-Sized Coreset

Robust query optimization: given a workload distribution W and plan space Π , select a small set of plans whose risk *covers* the workload, forming a coreset that supports local plan inference.

Lemma 0.51 — greedy termination. Assuming the concentration conditions of the sampling lemmas hold in every round, the greedy cover terminates with $k_r \leq 2\kappa^*$.

Theorem 0.52 — risk coreset via greedy cover. Let W be a workload distribution. Given plans Π with $|\Pi| = N$ and parameters $\varepsilon \in (0, 1]$, $\tau \geq 1$, $\delta \in (0, 1)$, let κ^* denote the size of the smallest $(\varepsilon, \tau, \delta|W|)$ -coreset for risk. Fix any constants $c_1, c_2 > 1$. There exists an algorithm that samples $O((\kappa^*/\delta^2) \ln N \ln(\delta^{-1}))$ queries from W and computes, in time $O((\kappa^*/(\varepsilon^2 \delta^2)) \cdot N \ln^3 N \ln(\delta^{-1}))$, a set $C \subseteq \Pi$ of size $O(\kappa^* \ln \delta^{-1})$ such that, with high probability, C is a $(c_1 \varepsilon, \tau, c_2 \delta|W|)$ -coreset for risk.

This is the set-cover pattern in the wild: risk coverage is submodular, each greedy round shrinks the residual failure probability geometrically (the termination lemma caps rounds at twice the optimum), and the price of coverage is the familiar logarithmic slack $\ln(\delta^{-1})$ in the coreset size. Hoeffding and union bounds merely ensure the greedy cover runs on faithful proxy risks rather than true ones ^{84PODS}.

84

Kruskal Edge-Exchange: Exact Indexing for Temporal Graphs

Answering cohesive-subgraph queries (connected components, k -core) over *all* time windows of a temporal graph, from one spanning-forest index built Kruskal-style with amortized union-find.

Lemma 0.53 — affiliated-forest invariant. Given a spanning edge $e = (u, v, t_1, t_2)$, let $\tau(e)$ be the timestamp such that $e \in F_{t_1, \tau(e)}$ and $e \notin F_{t_1, \tau(e)+1}$. An affiliated forest $(V, F_{x,y})$ built by BuildAF contains e if and only if $0 \leq x \leq t_1$ and $t_2 \leq y \leq \tau(e)$.

Here the exchange is exact, not approximate: graphic-matroid cycles let the construction swap an arriving edge for an expiring one while keeping a forest valid for every window simultaneously—Lemma above is the invariant those exchanges preserve. The certificate is completeness, not a factor: an $O(m \log m)$ construction and $O(m)$ -space index answer any window exactly, with queries in $O(n + \log^2 m)$ ⁸⁵.

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Swap Exchanges for Symmetry Breaking in Exact Search

Exact branch-and-bound for maximum common induced subgraph (MCIS), NP-hard, with branches pruned using vertex and value symmetries.

Theorem 0.54 — soundness of symmetry breaking. Variable symmetry breaking does not violate the correctness or completeness of the MCIS solution.

The proof is a pure swap: exchanging the conflicting pairs $(v_1, u_2), (v_2, u_1)$ for $(v_1, u_1), (v_2, u_2)$ maps any pruned branch to a symmetric one considered *earlier* in the fixed branching order, so some optimal mapping always survives the pruning. The exchange partner is supplied by automorphisms—two vertices are modular-symmetric exactly when their transposition is an automorphism of the graph—turning group symmetry into a completeness certificate for exact search ⁸⁶.

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When to reach for exchange-greedy

- Your selection, covering, or clustering problem is proved NP-hard by reduction, and the objective has *checkable* structure: diminishing returns (submodularity), a covering constraint, metric distances, or matroid independence.
- You want the algorithm's quality to be a theorem, not a benchmark: the exchange proof is usually a page, and the factor $(1 - 1/e, H(n), 3 + \epsilon, 1/2, \text{ or exact})$ is inherited, not re-derived.
- Audit the hypotheses, not the algorithm: pairwise marginal

tests or a concave-gain decomposition for submodularity; the triangle inequality for metrics; the exchange axiom for matroids. Near-modular or noisy objectives void the certificate.

- $1 - 1/e$ needs *monotone* submodular; without monotonicity fall back to randomized variants ($1/2$ unconstrained).
- The same idea certifies *exactness*: forest exchanges and symmetry swaps prove that pruning or one-pass construction loses nothing—completeness with no approximation factor at all.

Amortized Analysis and Potential Functions

Amortized analysis is the systems community's favorite certificate that rebuilds, compression, merges, and splits are *free*: a single operation may detonate an $O(n)$ reorganization, but if many cheap operations must accumulate before the next detonation becomes possible, the cost per operation stays $O(1)$. Eleven papers in this corpus reach for the amortized/potential toolbox, and they own one of the most lopsided tool–property pairings in the survey: the pair *upper-bound* $\times$ *amortized-potential* occurs in ten of the eleven. The tool almost always arrives to certify a per-operation time or space bound that an honest worst-case statement could not quote.

The tool

Three equivalent framings ⁸⁷:

87

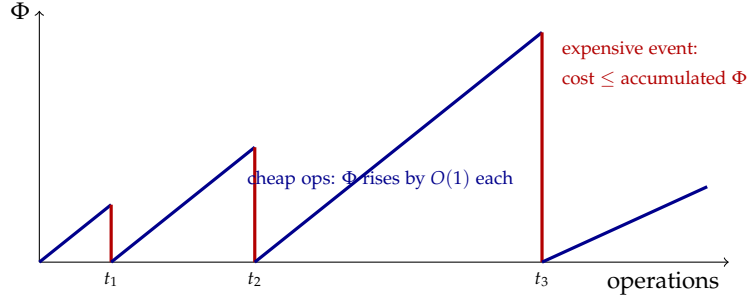
- **Aggregate method:** bound the total cost $T(n)$ of the worst sequence of n operations and report $T(n)/n$.
- **Accounting (banker's) method:** charge each operation a fictitious amortized price; cheap operations bank the surplus, expensive operations spend it, and the balance must never go negative.
- **Potential (physicist's) method:** fix a function Φ from data-structure states to $\mathbb{R}_{\geq 0}$ and define the amortized cost $\hat{a}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$. Telescoping gives $\sum_i c_i \leq \sum_i \hat{a}_i$ whenever $\Phi(D_n) \geq \Phi(D_0)$.

The design recipe for Φ has three canonical rules:

1. **Nonnegativity.** $\Phi(D) \geq 0$ for every reachable state (typically $\Phi(D_0) = 0$) — this is what makes the telescoping inequality safe for every prefix of the sequence.
2. **Deposits.** Every cheap operation raises Φ by at most its amortized cost: $+O(1)$ per insertion into a counter, a slack variable, or a size-excess budget.

3. **Withdrawals.** Every expensive event consumes a *large* accumulated Φ : it can fire only once some measurable quantity (counter, excess, rounding slack) has crossed a threshold, and Φ then drops by at least the event’s cost.

Figure 7: The sawtooth: cheap operations deposit into Φ ; an expensive event fires only when enough has accumulated, and its cost is paid entirely by the drop in Φ . Doubling the event cost doubles the inter-event gap.



Amortized bounds beat worst-case bounds exactly when expensive events are rare *because they require accumulation*: a worst-case bound must quote the burst cost C for every operation, while the amortized bound divides C by the $\Theta(\text{accrual})$ operations that had to precede it. The classic showcases are dynamic tables, Fibonacci heaps, and splay trees, where Φ is a sum of imbalance terms⁸⁸; periodic global rebuilding is the degenerate special case “rebuild cost over ℓ ” for period ℓ ⁸⁹. The signature recipe in this corpus is *rare-event bounding by a potential function*: a GK-style quantile sketch holds at most $2s \ln(n/s + 2)$ tuples because every compaction spends accumulated error slack⁹⁰, and a B-tree root split needs amortized $\Omega(b^{h+2})$ insertions before it can recur. The price of the method: the bound is a promise about *sequences*, not single operations — latency- or adversary-sensitive settings require deamortized variants.

Five views from the corpus

The five exemplars below exhibit progressively looser readings of the method: an explicit sawtooth potential, an implicit batching argument, potential as rounding slack, amortization imported as a black box, and potential as a progress measure.

SplineSketch: the sawtooth, literally

A streaming quantile sketch maintains k buckets over the keys seen so far, each with a counter b_i of absorbed items; arrivals increment counters, a bucket whose counter violates its budget ($b_i > C_b \cdot n/k$) must *split*, and adjacent buckets are *joined* to reclaim space. The maintenance question is how many splits a stream of n items can force.

⁸⁸; and

⁸⁹

⁹⁰

Lemma 0.55 — Split budget. Combining the lower bound on the decrease due to a split with the upper bound on the total potential, the number of splits during $(t', t]$ is at most

$$\frac{t}{\beta \cdot C_b \cdot \frac{t}{2.5k}} = \frac{2.5k}{\beta \cdot C_b} \leq \frac{k}{18},$$

where C_b is chosen based on β .

The potential is a function of the bucket counters — their excess over the budget $C_b \cdot n/k$: every arrival deposits $O(1)$, and the expensive event, a split, consumes a constant fraction β of what has accumulated, which is literally the sawtooth of Figure 7. The split budget then lets the rank error be charged along the split history, giving $\varepsilon^n(\sigma_i) \leq O(\log \alpha \cdot n/k)$ for any threshold σ_i of the final sketch ⁹¹.

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AeroSketch: batching the re-decomposition

Tracking a covariance sketch of a row stream requires repeatedly extracting the top eigenvectors of a $d \times d$ matrix — far too expensive to redo on every arrival.

Theorem 0.56 The algorithm solves the ATTP matrix-sketch problem with space complexity $O(d \log \|A\|_F^2)$ and amortized time complexity $O(d \log d)$ per update; with probability amplification it achieves amortized time complexity $O(d \log d \log^2 \frac{1}{\delta})$ per update while guaranteeing success probability at least $1 - \delta$, where δ is a tunable parameter.

There is no explicit Φ here: the expensive event is the eigen-decomposition of the accumulated covariance, executed by simultaneous iteration in $O(d \log d)$ using $\log_2 d + 1$ power-iteration steps, and it fires only when a dump threshold θ is crossed; each arriving row pays its share of the burst, i.e. the amortized cost is burst/ ℓ for batch length ℓ . A decomposability lemma — per-site sketches compose additively in covariance error — then carries the same accounting to sliding windows and distributed sites ⁹².

92

Approximate query processing under updates: slack as potential

For free-connex conjunctive queries over a semiring, exact incremental view maintenance is blocked by dynamic lower bounds outside the tiny q-hierarchical class. The escape is to keep auxiliary views only approximately: each cached annotation is rounded up within a multiplicative factor b and refreshed lazily when the true value outgrows the cache.

Theorem 0.57 For any $b > 1$ and any free-connex query $Q(H, F)$, all auxiliary views can be updated in amortized $O(\log_b M)$ time under an insertion-only update sequence, where $M = \max_{i,e} \max_{t \in R_e^i} A_{R_e}^i(t)$.

Applied to maintaining the join size of a 3-way join, which has $M = O(N^2)$, the amortized update cost is thus $O(\log_b N)$.

The potential is the total rounding slack $\sum_t \log_b A_R(t)$: an insertion deposits $O(1)$, and the expensive event — a cascading recomputation up the view hierarchy, triggered when a tuple’s annotation exceeds its cached value by the factor- b slack — consumes a unit, so each tuple can fire at most $\log_b M$ times. The insertion-only hypothesis is essential: fully-dynamic sequences face an amortized $\Omega(\sqrt{N})$ lower bound for non-q-hierarchical queries, and it holds even under approximation ⁹³.

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Subset sampling over joins: amortization imported

One-shot and dynamic subset sampling over an acyclic join Q must draw each result tuple $u \in \text{Join}(Q)$ independently with probability $p(u)$, where p factorizes over the relations of Q ; the index shards tuples into weight buckets — each sub-instance is either light or near-uniform — with a meta-index steering rejection sampling across buckets.

Theorem 0.58 Given an acyclic join instance Q and a set of associated weight functions $\{p_j^R\}_{j \in Q}$, we can maintain a dynamic index under tuple insertions using $O(N \log N)$ space, with amortized update time $O(\log^3 N \log \log N)$, while supporting the subset sampling query in $O(1 + \mu_\Psi \log N)$ expected time, where N is the total number of insertions.

The expensive events are index rebalancing and the periodic re-derivation of per-bucket statistics as insertions shift tuples across weight classes; the amortized $O(\log N)$ maintenance primitive is imported rather than re-proved (the paper’s Lemma 5.1, from prior work: the direct-access join index over real and dummy tuples updates in amortized $O(\log N)$ per inserted tuple). The composability lesson: a data structure with an amortized guarantee can serve as a black-box component inside a larger amortized analysis ^{94PODS}

94

Jaguar: potential as progress measure

Jaguar evaluates conjunctive queries in submodular-width time by recursively decomposing the instance; at each node, CALIBRATE re-derives size bounds $g(Z)$ for subsets Z of the query variables — in fact a shortest-path computation on a weighted graph whose vertices are the subsets of V .

Theorem 0.59 Let $\varepsilon > 0$ be an arbitrary constant. Given a CQ Q , a set of statistics (Σ, n) , and a database instance D_0 that matches the schema of Q and satisfies the statistics (Σ, n) , Jaguar correctly computes $Q(D_0)$

in time $O(N^{\text{subw}(Q, \Sigma, n) + \varepsilon} + \text{OUT})$ in data complexity, where $N = |D_0|$ and OUT is the output size. In particular, after a preprocessing time of $O(N^{\text{subw}(Q, \Sigma, n) + \varepsilon})$, Jaguar supports constant-delay enumeration of the output tuples.

Here the potential is not a bank but a *progress measure*: the log-relation-size bound g . Light recursion edges leave the statistics intact (a light path has length at most $2^{|V|} = O(1)$), while each heavy edge decreases the potential by at least ε — so any root-to-leaf path carries at most $O(1)$ heavy edges and the recursion tree has $O(1)$ depth in data complexity. The expensive event is the heavy, statistics-tightening recursive call: it can fire only while potential remains to burn, and with linear-time CALIBRATE plus the invariant $c \leq \text{subw}(Q, \Sigma, n)$ on every calibrated bound, no level of the constant-depth tree exceeds the advertised $N^{\text{subw}(Q, \Sigma, n) + \varepsilon}$ 95PODS

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When to reach for amortized analysis

- You need a *per-operation* time or space bound, and the algorithm has an occasional expensive structural event — split, merge, rebuild, compression, re-decomposition, full scan — that only many cheap operations can set up.
- The trigger is measurable: some counter, excess, or slack that rises by $O(1)$ per cheap operation and drops sharply when the event fires. If you can write it as a Φ obeying the three rules, the bound follows by telescoping — and you get to tune the regime (batch length ℓ , slack ratio b , bucket budget C_b) to trade burst size against frequency.
- Sequences, not single operations: if the deployment cares about tail latency or faces an adversary who inspects timings, budget for deamortized variants; and remember that insert-only analyses can shatter under deletions (the $\Omega(\sqrt{N})$ fully-dynamic lower bound above).

Spectral and Matrix Tools

Why do eigenvalues keep appearing in database papers? Because the field's core objects are linear-algebraic in disguise: similarity and embedding quality are spectral properties of affinity and covariance matrices; the structured inverses arising from random walks, kernels, and fixed-point equations exist—and can be approximated—precisely when their spectra stay clear of zero; iterative processes converge at rates set by spectral radii and eigengaps; and low-rank summaries are optimal because of singular values. In this corpus, 13 papers reach for this machinery; after a compact toolbox we see three exemplars using it three different ways.

The tool: workhorse results

Throughout, A is symmetric with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$ and singular values $\sigma_1 \geq \sigma_2 \geq \dots$; all statements are classical (see ⁹⁶; for graphs specifically ⁹⁷).

⁹⁶; and
⁹⁷

Proposition 0.60 — Courant–Fischer.

$$\lambda_k(A) = \min_{\dim V = n-k+1} \max_{0 \neq x \in V} \frac{x^\top A x}{x^\top x} = \max_{\dim V = k} \min_{0 \neq x \in V} \frac{x^\top A x}{x^\top x}.$$

Hence if $B \in \mathbb{R}^{n \times s}$ has orthonormal columns, the eigenvalues μ_i of the compression $B^\top A B$ interlace: $\lambda_i(A) \geq \mu_i \geq \lambda_{n-s+i}(A)$ (Poincaré separation).

Lemma 0.61 — Weyl; Davis–Kahan $\sin \Theta$. $|\lambda_k(A + E) - \lambda_k(A)| \leq \|E\|_2$ for symmetric E ; and an invariant subspace U of A separated from the rest of the spectrum by a gap Δ moves to $\tilde{U}$ of $A + E$ by at most $\|\sin \Theta(U, \tilde{U})\|_2 \leq \|E\|_2 / \Delta$: subspace stability is perturbation norm over eigengap.

Theorem 0.62 — matrix Bernstein ⁹⁸. For independent, centered, symmetric $d \times d$ matrices Y_i with $\|Y_i\|_2 \leq R$ surely and $\sigma^2 = \|\sum_i \mathbb{E} Y_i^2\|_2$,

$$\Pr \left[\left\| \sum_i Y_i \right\|_2 \geq t \right] \leq 2d \exp(-t^2 / (2(\sigma^2 + Rt/3))).$$

This is the bridge to the concentration chapter: sums of independent matrices concentrate like sums of scalars, paying only a factor d .

⁹⁹; and

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¹⁰¹

Also worth pocketing: by Eckart–Young–Mirsky, truncated SVD is the optimal rank- k summary ($\|A - B\|_2 = \sigma_{k+1}$, $\|A - B\|_F = (\sum_{i>k} \sigma_i^2)^{1/2}$); power iteration contracts the angle to the dominant direction by σ_2/σ_1 per step (squared on $A^\top A$; $(\sigma_{k+1}/\sigma_k)^q$ over q subspace-iteration steps), so the gap is the convergence rate ⁹⁹. Eigenvalues of a Kronecker sum are pairwise sums, $\lambda(A \oplus B) = \{\lambda_i(A) + \lambda_j(B)\}$, and $\rho(A) < 1$ gives the Neumann expansion $(I - A)^{-1} = \sum_{t \geq 0} A^t$ with geometrically decaying truncation error ¹⁰⁰; orthogonal projections are non-expansive. For fixed x , the first d rows of a random orthogonal matrix satisfy $\mathbb{E}\|Ax\|_2^2 = (d/D)\|x\|_2^2$ with chi-square tails, so $d = \Theta(\epsilon^{-2} \log(1/\beta))$ dimensions give $(1 \pm \epsilon)$ distortion (distributional Johnson–Lindenstrauss) ¹⁰¹.

Three views from the corpus

Effective clustering for large multi-relational graphs

An NP-hard normalized-cut-type objective over a dense multi-relational affinity is relaxed spectrally, with Sinkhorn–Knopp scaling and a Neumann series replacing a full eigendecomposition.

Theorem 0.63 — doubly stochastic affinity gives k -means. If S_ω is doubly stochastic and the relaxed embedding is $Z = Y^\top$, then optimizing the relaxed objective is equivalent to optimizing

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{v_i \in C_k} \|z_i - c^{(k)}\|_2^2, \quad c^{(k)} = \sum_{v_j \in C_k} \frac{y_j}{|C_k|},$$

where $c^{(k)}$ is the center of cluster C_k .

¹⁰²

The spectral work sits in producing Y : a truncated Neumann series $(I - Z)^{-1} = \sum_{t \geq 0} Z^t$, valid since the dominant eigenvalue obeys $|\lambda| < 1$, with truncation error after L terms decaying geometrically (on the order of $(\alpha/(1+\alpha))^{L+1}$) at a rate governed by an eigengap; the supporting lemmas are that expansion, the Weyl/Lidskii-type inequality $\lambda_{\max}(A + B) \leq \lambda_{\max}(A) + \lambda_{\max}(B)$ for Hermitian matrices, and a CountSketch estimate of residual Frobenius norms. ¹⁰²

Subspace collision for high-dimensional approximate nearest neighbor search

k -ANNS by counting “collisions” of points with the query across many s -dimensional subspaces needs projections that preserve neighbor orderings.

Theorem 0.64 — ordering preservation. Let $B = [B_1 \dots B_{N_s}] \in \mathbb{R}^{d \times (N_s \cdot s)}$ with $N_s \cdot s \leq d$ have orthonormal columns, i.e. eigenvectors of the covariance, so that $BB^\top$ is an orthogonal projection. Suppose the local difference vectors satisfy $\|(I_d - BB^\top)(o_i - o_j)\|^2 \leq \varepsilon \|o_i - o_j\|^2$ for some $\varepsilon \in (0, 1)$. Then $\|B^\top(o_i - o_j)\|^2 \geq (1 - \varepsilon) \|o_i - o_j\|^2$. Consequently, whenever $\|o_i - o_j\|^2 < (1 - \varepsilon) \|o_i - o_z\|^2$, the ordering survives the transformation: $\|B^\top(o_i - o_j)\| < \|B^\top(o_i - o_z)\|$.

The load-bearing facts are Pythagoras and interlacing: the lower bound is the decomposition $\|x\|^2 = \|B^\top x\|^2 + \|(I - BB^\top)x\|^2$ plus the hypothesis, the upper bound the non-expansiveness of orthogonal projections. Design is settled by Poincaré separation: since the eigenvalues of each compressed covariance $B_j^\top \Sigma B_j$ interlace those of Σ , monotonicity of the logarithm turns log-determinant subspace allocation into an eigenvalue assignment. ¹⁰³

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Weighted Fourier factorizations for private marginal queries (PODS)

Under μ -GDP Gaussian noise, all marginal queries induced by a collection $\mathcal{S}$ of subsets of $[d]$ must be answered, with weights and a factorization chosen to minimize weighted error.

Theorem 0.65 — weighted factorization attains $\gamma_{2,F}$. With workload matrix W for the marginal workload $Q_{\mathcal{S}}$ over $U = U_1 \times \dots \times U_d$, $U_i = \{0, \dots, m_i - 1\}$, weights $p : \mathcal{S} \rightarrow \mathbb{R}_{\geq 0}$, and P the diagonal matrix having entries $P_{(S,t),(S,t)} = p(S)/|U_S|$ for $t \in U_S$, the factorization $LR = W$ of the paper’s Algorithm 1 satisfies

$$\gamma_{2,F}(PW) = \|PL\|_F \|R\|_{1 \rightarrow 2} = \left(\sum_{R \in \mathcal{S}^\downarrow} \sum_{\substack{T \in \mathcal{S} \\ T \supseteq R}} \sqrt{\sum_{j \in R} \frac{p(T)}{|U_T|} (m_j - 1)^2} \right)^{1/2},$$

where $\mathcal{S}^\downarrow$ is the downward closure of $\mathcal{S}$. Moreover, for the weights p^* maximizing the right-hand side over the simplex $\{p \geq 0 : \sum_S p(S) = 1\}$, the resulting mechanism is μ -GDP for all of $Q_{\mathcal{S}}$, its worst-case error equals $1/\mu$ times the same expression at p^* , and this bound is optimal among all factorization mechanisms.

Nothing here is spectral in the eigenvalue sense—the machinery is *factorization norms*: the error of a Gaussian-noise factorized release is governed by $\|PL\|_F \|R\|_{1 \rightarrow 2}$, identified with the factorization norm $\gamma_{2,F}$; first-order (KKT) conditions on the weight LP pin p^* , and Gaussian concentration lifts average error to worst-case error. ¹⁰⁴PODS

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When to reach for spectral tools

- **Relaxing a combinatorial objective.** Eigenvectors of the (Sinkhorn-balanced, doubly stochastic) affinity turn NP-hard partitioning into k -means on an embedding; Neumann series and eigengaps control the approximation.
- **Certifying invertibility of a structured matrix.** For Kronecker sums and $I - Z$, localize the spectrum (pairwise sums; spectral radius below one); positivity or contraction yields existence, geometric approximation, convergence rates.
- **Choosing or defending a subspace.** Interlacing says compressions inherit sandwiched eigenvalues—use it to optimize what to project onto; Davis–Kahan bounds the drift of the chosen subspace by perturbation over eigengap.
- **Making distances cheap.** Random orthogonal projections are approximate isometries with chi-square tails (distributional JL), so projected thresholds separate near from far with controlled error probabilities.
- **Bounding error of a noisy linear mechanism.** When the strategy is a factorization $LR = W$, error is a product of operator norms ($\|PL\|_F \|R\|_{1 \rightarrow 2} = \gamma_{2,F}$); when noise is a sum of independent random matrices, matrix Bernstein takes over.
- **Caveats.** Weyl, Davis–Kahan, and interlacing need symmetric structure—otherwise argue about singular values or a Schur form; eigengap hypotheses are load-bearing; JL dimensions $d = \Theta(\epsilon^{-2} \log(1/\beta))$ decide whether projecting is actually cheaper.

Online decision-making: competitive analysis and regret

Online decision theory enters this corpus through systems doors: when to adapt a filter to a recurring false positive, whether to trust an ML oracle’s prediction, how to clean a time series that has not yet arrived, how to tune a discretization knob. Four papers carry it—two via *competitive analysis*¹⁰⁵, two via *regret*¹⁰⁶, none an online-algorithms paper per se. They share one shape: the system commits to decisions while the input’s future is unknown, and the committed sequence is judged against a hindsight optimum that sees everything. Two classical yardsticks quantify that gap, and choosing between them is this chapter’s lesson.

¹⁰⁵ ; and

¹⁰⁶ ; and

The tools

Definition 0.66 (Competitive ratio). An online algorithm A for a cost-minimization problem is c -competitive if there is a constant b such that

$$\text{cost}_A(\sigma) \leq c \cdot \text{cost}_{\text{OPT}}(\sigma) + b \quad \text{for every input sequence } \sigma,$$

where OPT is the offline optimum that observes all of σ —including where it ends—before acting.

The canonical micro-example is *ski rental*: skis rent for 1 per day and cost B to buy; each morning the skier learns only whether the season continues. Renting for B days and then buying is 2-competitive, and no deterministic algorithm does better; randomization lowers the barrier to $e/(e-1) \approx 1.58$ against an oblivious adversary¹⁰⁷. The framework descends from the amortized analysis of list update and paging¹⁰⁸. Figure 8 shows the geometry; the memoryless “when do I commit?” structure is why this toy model reappears below in database clothing.

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Definition 0.67 (Regret). For decisions $a_1, \dots, a_T$, where round t charges a loss $\ell_t(a_t)$ revealed only after the decision, the *regret* against the best

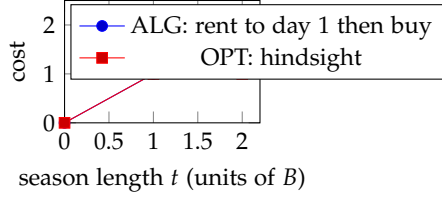


Figure 8: Ski rental with rent 1 and buy price $B = 1$: rent-to- B -then-buy matches OPT while $t < B$ and never exceeds $2 \cdot$ OPT.

fixed comparator is

$$R_T = \sum_{t=1}^T \ell_t(a_t) - \min_{a \in \mathcal{A}} \sum_{t=1}^T \ell_t(a).$$

An algorithm is *no-regret* if $R_T = o(T)$: its average loss converges to that of the best action chosen once, in hindsight.

In the stochastic multi-armed bandit (K actions, rewards in $[0, 1]$ with fixed unknown means, only the chosen arm’s reward observed), UCB plays the arm with the highest optimism index $\bar{r}_k + \sqrt{2 \ln t / n_k}$ and achieves $R_T = O(\sqrt{KT \log T})$ ¹⁰⁹; adversarial and unified views are surveyed in¹¹⁰. In short: the *competitive ratio* prices the unknown future—unrestricted adversary, switching comparator, guarantee on every input—for one-shot commitments under adversarial workloads; *regret* prices ignorance of a distribution—repeated rounds, per-round feedback, fixed comparator, $O(\sqrt{T})$ bound—for stochastic tuning loops.

Four exemplars from the corpus

Ski rental inside a sketch filter

A streaming distinct-count filter occasionally yields a false positive; each repeat costs a database lookup C_{DB} , adapting the entry costs an extra reverse-map lookup $C_{RM} = \alpha C_{DB}$, and after adapting both costs stop. Whether and when to adapt is ski rental in disguise—rental price C_{DB} , purchase price $C_{DB} + C_{RM}$, adversarial arrival stream.

Theorem 0.68 — Adapt after $\lfloor \alpha \rfloor$ repeats; 2-competitive. If $\alpha \geq 1$, i.e., the cost of querying the reverse map is greater than a database lookup, the optimal deterministic strategy is to adapt immediately after the false positive repeats $\lfloor \alpha \rfloor$ times, and is 2-competitive.

Remark 0.69 (The golden-ratio threshold). For $\alpha < 1$ the two candidate policies have ratios $1 + \alpha$ (adapt immediately) and $(2 + \alpha)/(1 + \alpha)$ (adapt on the second false positive); these cross where $\alpha^2 + \alpha - 1 = 0$, i.e. at $\alpha = (\sqrt{5} - 1)/2 \approx 0.618$. The optimal rule: adapt immediately when $\alpha < 0.618$, adapt on the second occurrence otherwise.

The competitive ratio is the right yardstick: the arrival stream is adversarial and the offline comparator knows when the repeats end. The

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¹¹⁰

skeleton also composes—augmenting a strongly adaptive filter with this routine bounds false positives by $(1 + \lceil \alpha \rceil)(n\epsilon + O(\sqrt{n\epsilon \log n + \log n}))$ with high probability, the 2-competitive core living inside a high-probability bound. ¹¹¹

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Paying for oracle predictions at enumeration time

Subgraph-isomorphism (SUBISO) enumeration is accelerated by an ML oracle that prunes the search at prediction cost $f(Q, M)$, but a wrong prediction misses matches. The paper’s β -robustness—extra output cost relative to the optimal exact enumerator—is a competitive ratio; the general case is barred, then per-algorithm ratios are proved.

Theorem 0.70 — No constant robustness in general. No algorithm exists for SUBISO that is both output polynomial and β -robust for any positive constant β unless $\text{fewP} = P$, where fewP denotes the class of problems recognizable by a nondeterministic Turing machine in polynomial time with only polynomially many solutions.

For the paper’s oracle-guided hybrids the matching upper bounds are constant competitive ratios, e.g. $5N_c(f(Q, M) + 2|G||Q|)$ for VF3_M with probability at least $1 - \epsilon$, proved via a generalized Bonferroni inequality over the oracle’s false predictions. The barrier is a parsimonious reduction; the ratios fit competitive analysis since the graph is arbitrary and the comparator is the exact enumerator’s cost—paid in oracle calls and output quality rather than money. ¹¹²

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Windowed time-series repair with decay regret

Given a multivariate time series violating soft and hard constraints, the offline optimal repair is computed exactly as the maximum-weight path in a DAG over anchor points. The online algorithm is the same dynamic program restricted to a sliding window of width W , so its regret is precisely what the truncated lookback misses.

Theorem 0.71 — Exponentially decaying window regret. The regret R_n of the online algorithm is bounded:

$$R_n = \sum_{i=1}^n r_i \leq \sum_{i=1}^n e^{-\beta(W+1)} = n e^{-\beta(W+1)},$$

where each one-step regret r_i compares the score of the offline optimal anchor point with that of the best anchor visible within the window, and the strictly decreasing path weights obey $\gamma(\Delta) \leq e^{-\beta\Delta}$.

The comparator is a best *sequence* (the exact offline optimum), a competitive-analysis comparator, yet the guarantee is a sum of per-step gaps in the regret style; the exponential decay makes a bounded

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window affordable—each extra unit of window multiplies the neglected tail by $e^{-\beta}$.¹¹³

Attribute discretization as a bandit

Choosing how to bin numerical attributes is posed as maximizing $J(B, D) = \alpha M_{\text{sem}}(B) + (1 - \alpha) M_{\text{util}}(B, D)$ over partition sets, with the search organized as an MDP over clusters of similar partitions. The only provable guarantee comes from the case where the search collapses to a repeated choice among K cluster-arms.

Theorem 0.72 — The single-attribute case is exactly a K -armed bandit. If (i) $d = 1$ (single attribute, one action per episode) and (ii) MCC uses the UCB index for action selection and updates, then Algorithm 1 becomes a UCB-guided MCC variant and reduces exactly to a stochastic multi-armed bandit with K arms (one per cluster) and reward in $[0, 1]$ given by J ; this equivalence does not extend to the multi-attribute case ($d > 1$). Running Algorithm 1 for T episodes yields the regret guarantees inherited from UCB: gap-free regret $O(\sqrt{KT \log T})$.

This is the reduction school of regret usage: engineer the system so an existing bandit theorem applies verbatim and inherit $O(\sqrt{KT \log T})$ for free. Regret fits because episodes repeat with stochastic rewards in $[0, 1]$ and the comparator is the best fixed cluster; for $d > 1$ the equivalence breaks and no guarantee is claimed.¹¹⁴

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When to reach for competitive analysis vs regret

- **Competitive ratio:** the decision is a commitment (adapt, buy, rebuild, trust an oracle) under a possibly adversarial input, and you need a guarantee on *every* run—ski-rental-shaped trade-offs between a repeated small cost and a one-time large one. Deterministic bounds are typically 2; randomization usually improves them once an adversary model is fixed.
- **Regret:** the same decision repeats over many rounds with per-round feedback and tacit stochasticity—tuning, configuration search, repair over stationary error processes. The comparator is a *fixed* action in hindsight; the $O(\sqrt{T})$ bound is weak per run, strong on average.
- **Hybrids are the corpus pattern:** reduce the system to a K -armed bandit and inherit UCB's bound wholesale; wrap a 2-competitive sub-routine inside a high-probability filter bound; measure a windowed online heuristic against the exact offline DP with a decay argument.

- **Check before you cite:** regret needs normalized rewards (here $[0, 1]$) and a defined feedback model; a competitive ratio needs a well-defined offline optimum and, if randomized, an oblivious adversary.